

Principles of modern LAN design and operation

Tecnologie e Servizi di Rete
Computer Network Technologies and Services

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*part of the slides adapted from material by Fulvio Rizzo



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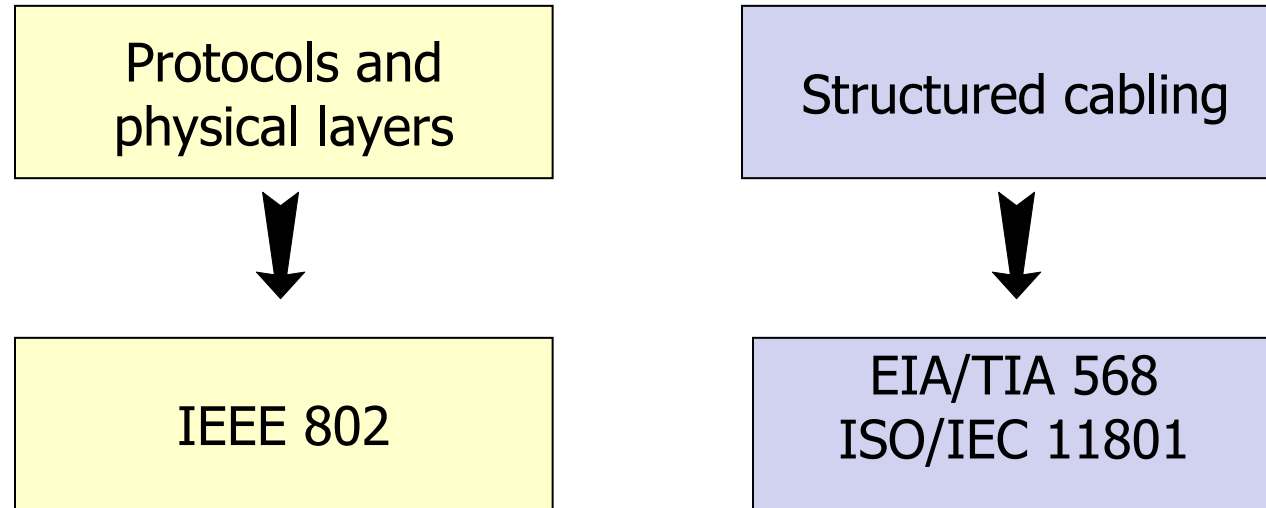
A view from history (1)

- Wide Area Networks appeared first
 - '60s
 - A few mainframes; necessity to connect to them from remote
 - Partition their expensive cost between more entities
- Local Area Networks appeared later
 - End 70's, beginning '80s
 - Minicomputers (and later PCs appeared)
 - Cost was low enough so that it was no longer needed to access a remote mainframe
 - Sharing resources between small workgroups (e.g., departments)
 - Mainframe still used, but for different purposes (e.g., scientific simulations)

A view from history (2)

- At the beginning, WAN and LANs evolved independently
- Different protocols, engineered by different vendors for different purposes
 - Decnet, SNA, IP
 - Novell, Banyan Vineis, NetBeui
- Later we tried to connect LANs to WANs
 - One winner: IP

LAN important standards

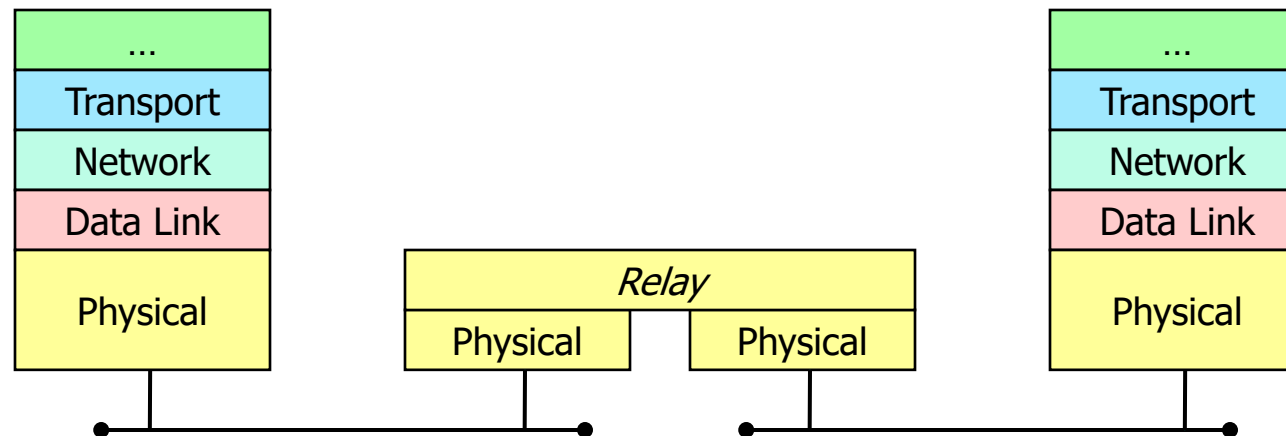


LAN devices in brief

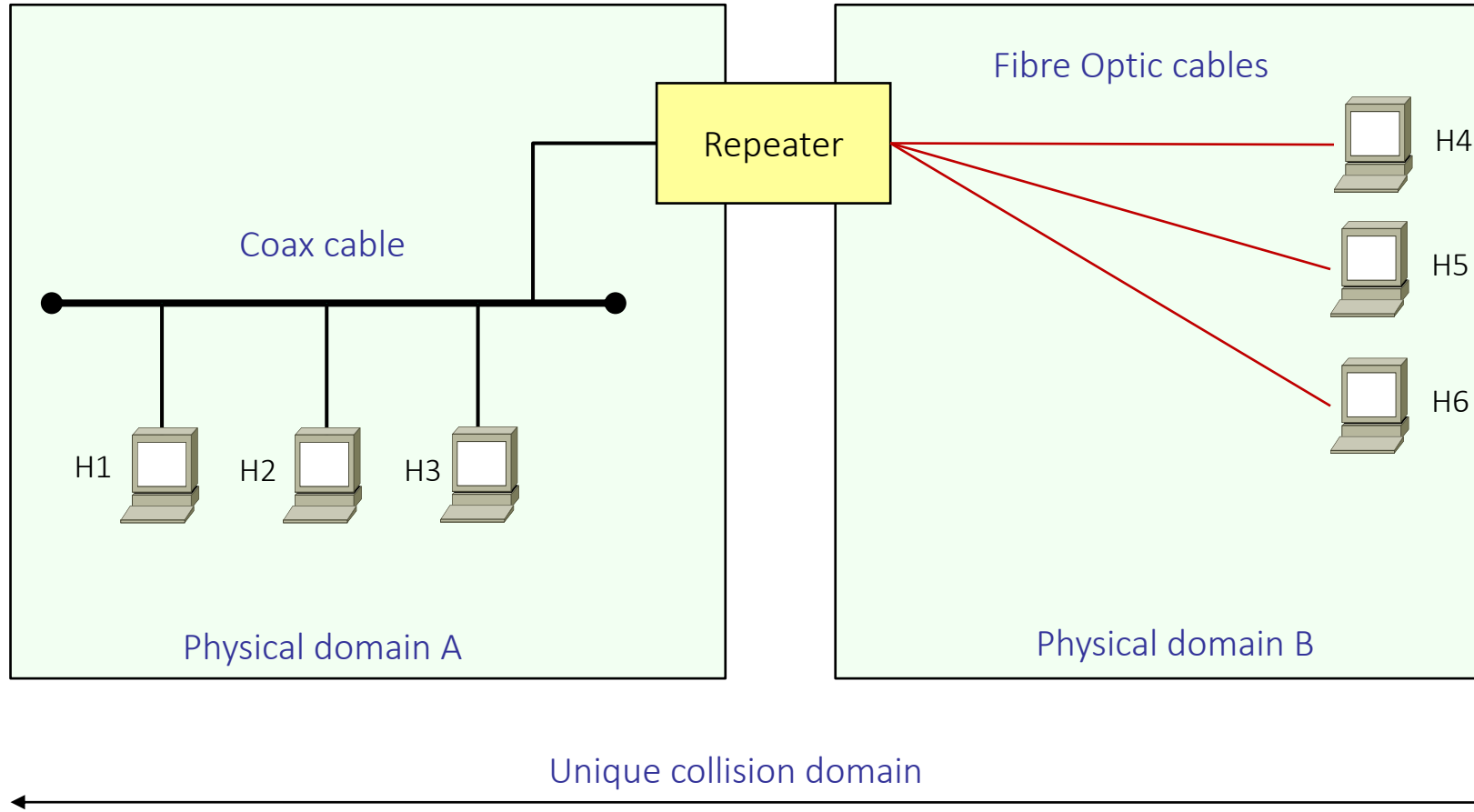
- L1: Repeater
 - Hub
 - Separate physical domains, same collision domain
- L2: Bridge
 - Switch
 - Separate collision domains, same broadcast domain
- L3: Router
 - L3 switch
 - Separate broadcast domains
 - Not really specific for LANs
 - Not covered in the current slides

Repeater

- Interconnection at the physical layer
 - Receives and propagates a sequence of bits
- Used for
 - Interconnecting networks having the same Medium Access Control (MAC)
 - Same technology but different physical layer is possible
 - Recovering signal degradation (long cables), allowing larger distances

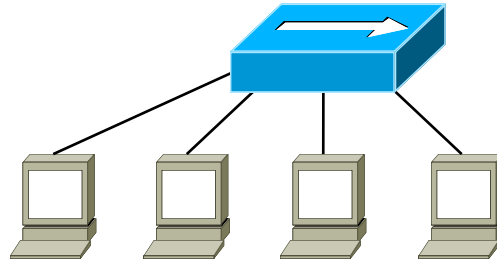


Repeater: example



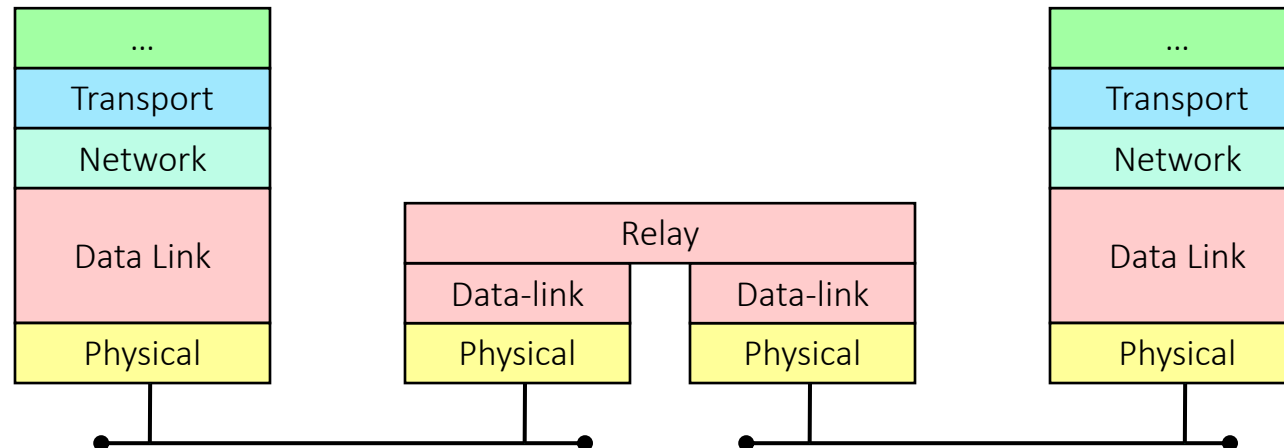
Multiport repeaters: Hubs

- Hubs are multiport repeaters
 - Repeaters with more than 2 ports
- Required for twisted pairs and fiber cabling (hub-and-spoke topology)



Bridge (1)

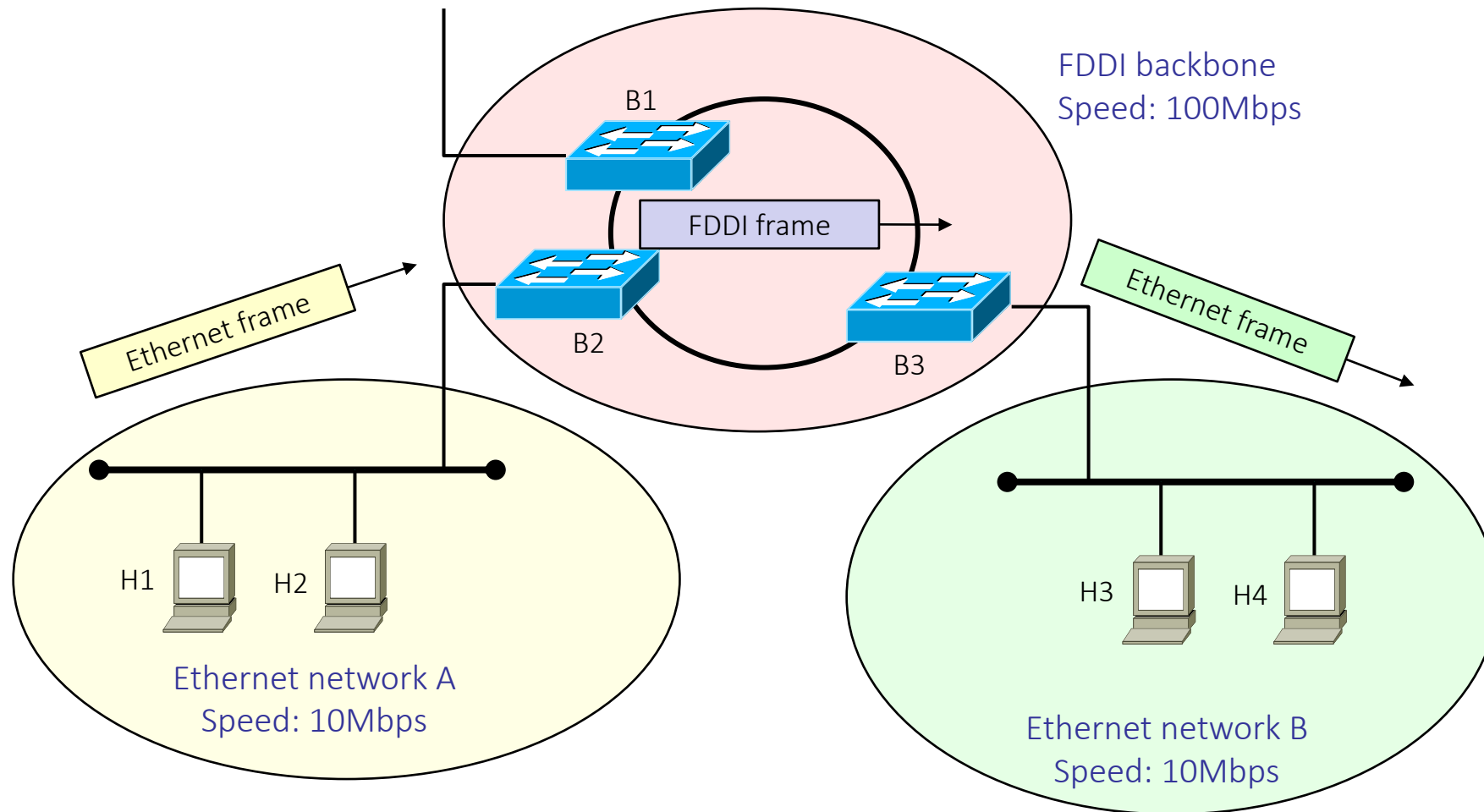
- Introduced by DEC in 1983 (LANBridge 100)
 - Pure software
 - 2 ports (mainly for economic reasons)
- Interconnection at the data-link layer
 - E.g., Ethernet to WiFi, Ethernet to Fast Ethernet
 - Different MACs (medium access mechanism, framing)



Bridge (2)

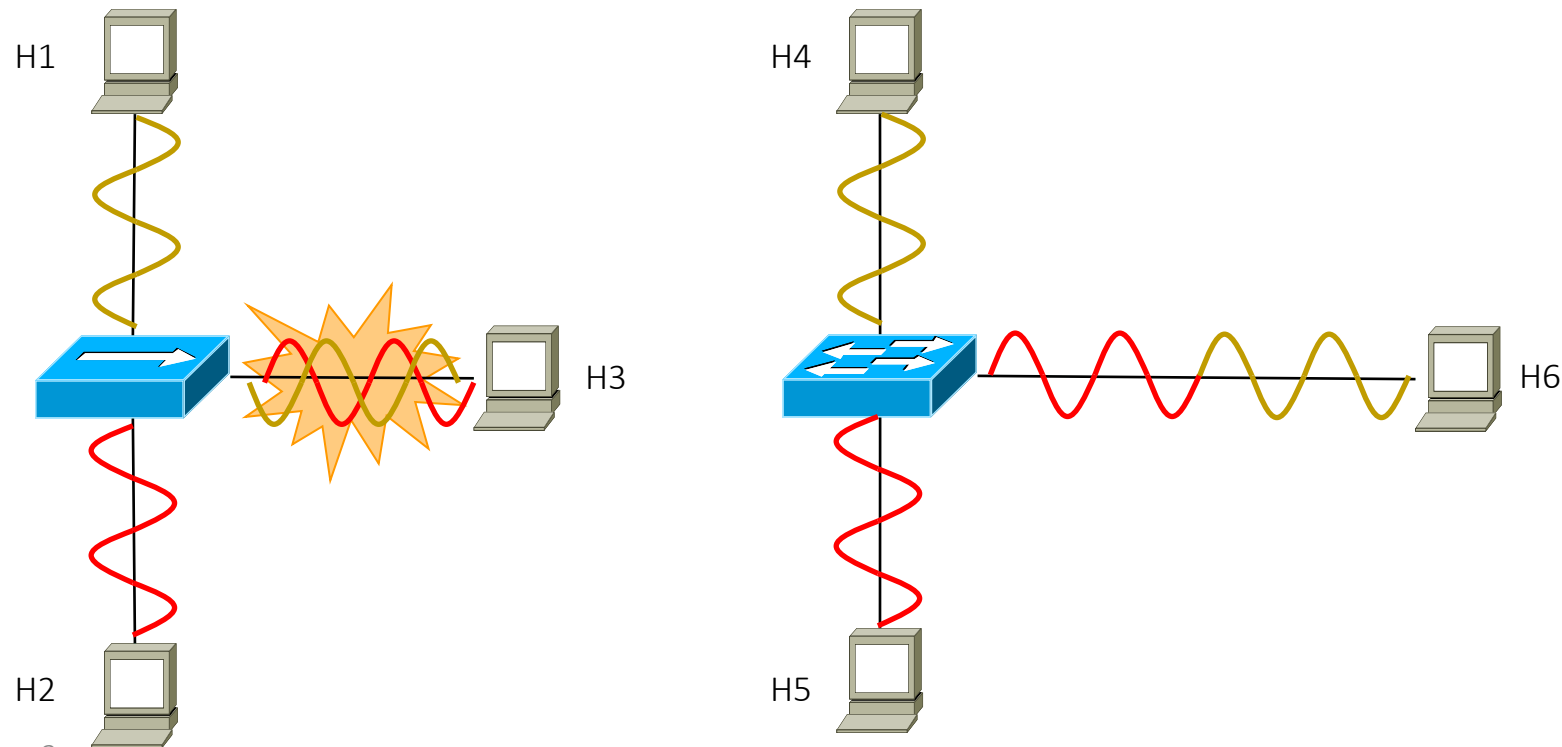
- LAN extension (total diameter)
 - Especially useful for FastEthernet and upper speed (200m)
 - Collision domain issues
- Works by receiving and re-transmitting (later) a frame
 1. Store the frame (*store and forward* mode)
 2. Modify the frame (e.g., Ethernet to Token Ring)
 3. Send it out

Bridge: example of interconnection



Bridges and collisions (on Ethernet)

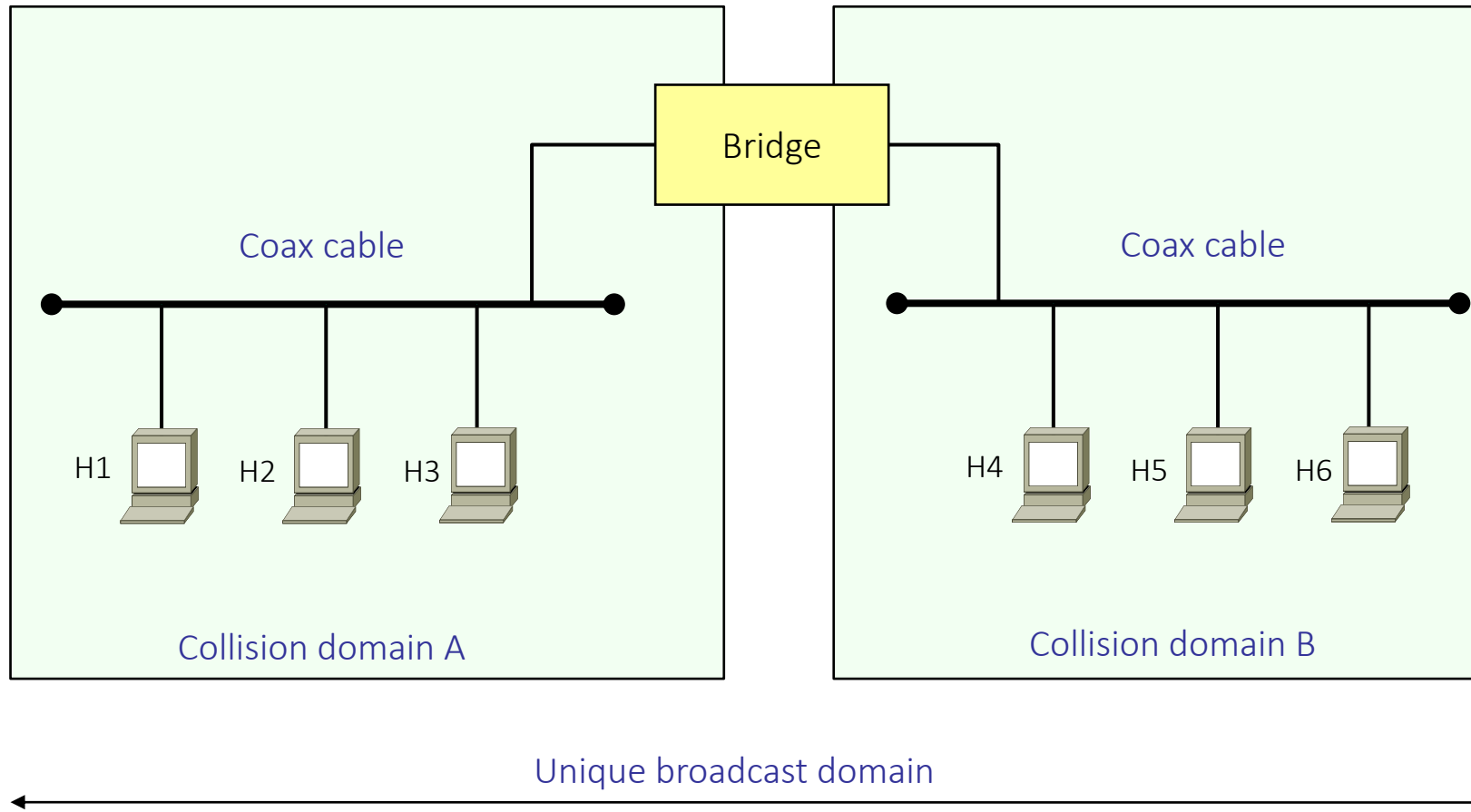
- “Store and forward” allows smarter sending of data on output interfaces
- Bridges **decouples collision from broadcast domain**
 - Collision domain is no longer a limitation



Collision and broadcast domains (1)

- **Collision domain:** area where a single instance of the access control algorithm (e.g., CSMA/CD) operates
 - I.e., the area covered by a *single “physical” link*
 - Frames are immediately propagated over all the links (possibly through repeaters)
 - Also called **network segment**
- **Broadcast domain:** area where frames can be propagated
 - I.e., the area on which a *LAN* operates
 - Can include several collision domains
 - Frames can be stored and later propagated over other collision domains

Collision and broadcast domains (2)



Collision and broadcast domains (3)

- Repeaters “extend” the collision domain
 - In fact, it is not actually “extended”; it allows the collision domain to reach its theoretical limits, despite cable limitations
- Bridges **create different collision domains** and **extend** the broadcast domain
 - I.e., bridges decouple broadcast domain from collision domain
- This is a very important feature of bridges, that comes out from their “store and forward mechanism”

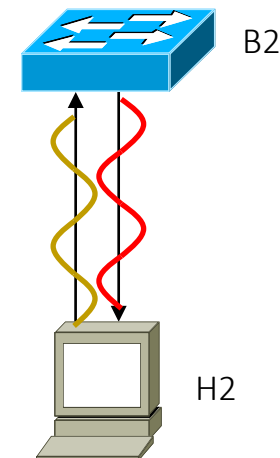
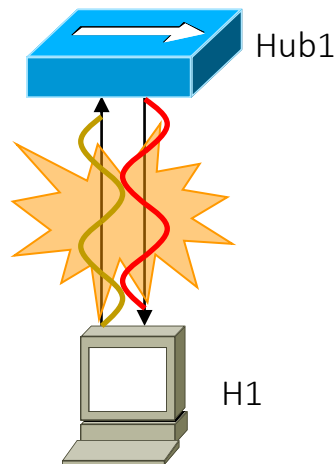


Half and Full duplex mode

- Half Duplex mode

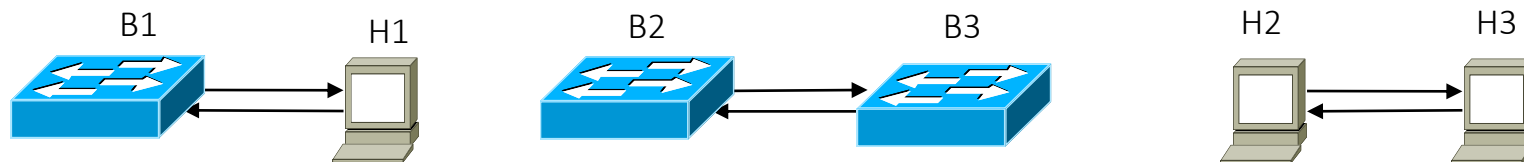
- Standard operating mode of network interfaces (NICs)
- RX and TX cannot happen at the same time
- RX+TX activity is seen as collision (e.g., in 802.11)

- But... if we have two physical links, do we really have a collision?



Full duplex

- Introduced with Fast Ethernet (part of 802.3x)
- Available whenever *the other party* can temporarily store the frame, instead of repeating (immediately) the received bits on the other ports, like a repeater does
- Simultaneous TX and RX are possible
- Not just host $\leftrightarrow$ bridge
 - Examples: host $\leftrightarrow$ host, host $\leftrightarrow$ bridge, bridge $\leftrightarrow$ bridge



Full duplex: advantages

■ Bandwidth

- In theory, throughput x2
- In practice, limited advantage for clients and servers
 - Clients tend to saturate downlinks, servers uplinks
- May be interesting for bridges on the backbone
 - More symmetrical bandwidth

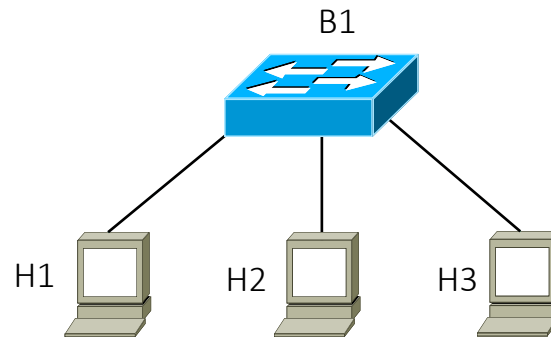
■ CSMA/CD: the real advantage

- No longer needed, since collisions are no longer possible
 - With CSMA/CD, TX and RX together are used to detect collisions
- Advantages
 - No requirement for min frame size for Ethernet
 - No limits on the network size on Ethernet (no collision domain)



Full duplex and switches

- Modern LANs are heavily based on full duplex
- Hub-and-spoke topology
 - Point-to-point connections between hosts and the “bridge”
 - No collision domain
- Multiport bridges are called “switches”
 - Same functions, different internal architecture



Modern LANs: Switched Ethernet

- Modern (wired) LANs are based on full-duplex, switches, and Ethernet:
Switched Ethernet
 - Today Gigabit Ethernet, or even more (10GE, 100GE)
- If today we say “switch”, we are referring to an Ethernet switch
- CSMA/CD no longer used
 - Available till 1GE, then no longer defined by standards
- Wireless LANs are completely different
 - Typically based on CSMA/CA (e.g., WiFi)
 - Hubs are still used (e.g., WiFi extenders)

...Here we focus on wired Ethernet-based LANs

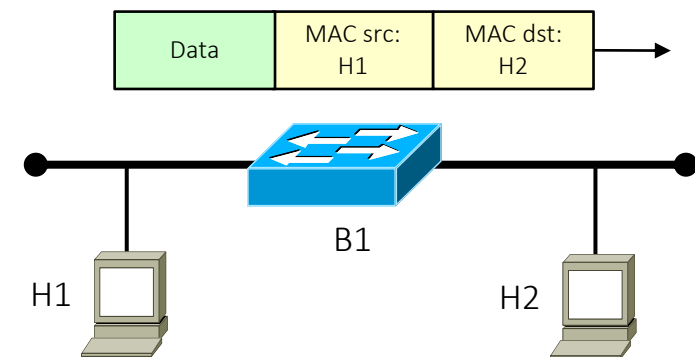
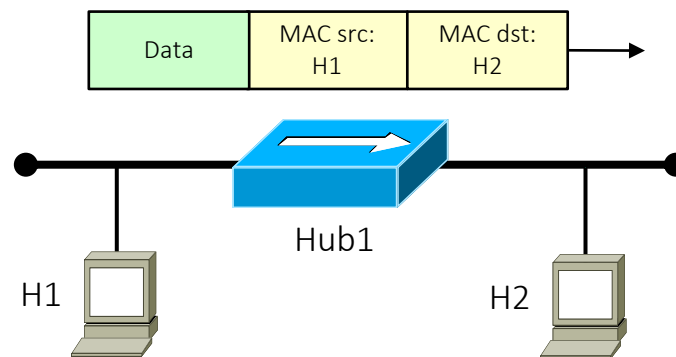
Transparent bridges/switches

- Bridges (switches) used in Ethernet LANs are called **transparent bridges/switches**
 - Other (non-transparent) bridges have been proposed in the past (e.g., Token Ring networks)
 - No longer in use
- Transparent bridges standardized by IEEE in 802.1D
- Transparency
 - Bridges (switches) should be plug-and-play and must not require any change in the configuration of end systems
 - Performance (throughput, max distances) may be different from the original network, but functionalities are the same



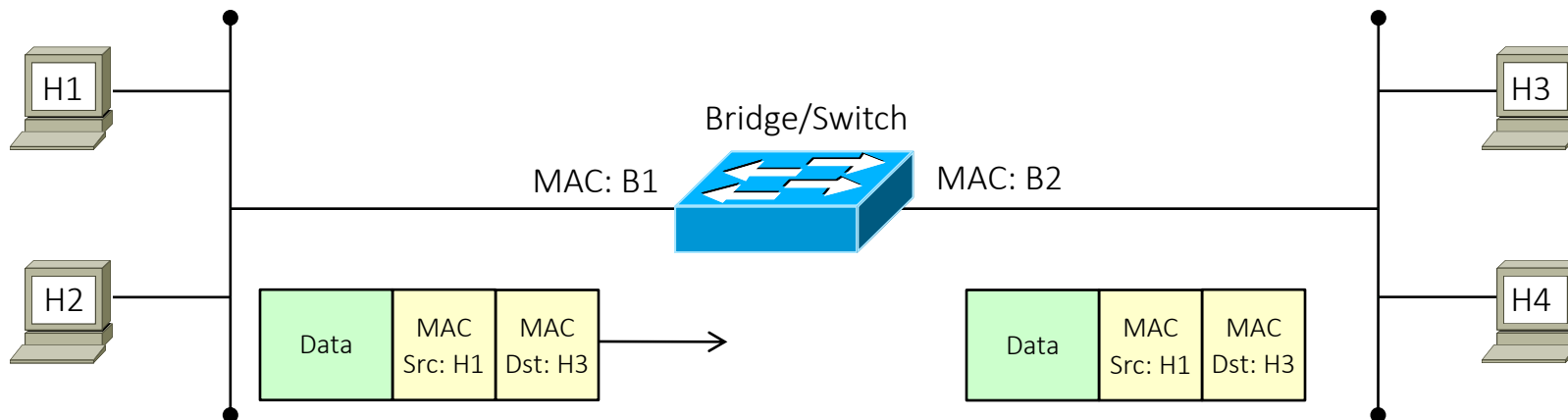
Transparent bridges/switches and end hosts

- End systems must operate in the same way (same frames, some format, etc) with or without bridges/switches
- In detail:
 - No changes at all in frames sent by end systems
 - Same frame, same src/dst MAC address, etc...
 - There may be some changes in **which** frames are received
 - No changes at all in the **format** of the received frame
 - Same source/MAC address, etc



Transparent bridges/switches and port addresses

- Each port of a bridge/switch has a MAC level and therefore it has a MAC address
 - That MAC address is never used when forwarding data frames
 - It is used when frames are generated/received by the switch itself
 - E.g., management frames



Smart forwarding process

- Smarter forwarding rules
 - Unicast: only on the port toward the destination (Destination MAC-based forwarding)
 - Multicast, broadcast: flooding
 - All ports except the port on which the frame has been received (*flooding*)
- A MAC forwarding table must be available locally
 - Filtering database (more details later)
- A note about flooding
 - Frames are sent on all ports (except on the incoming port), but **may not be sent at the same time** (delayed forwarding)
 - Hubs flood data at the same time (*bits* forwarded immediately)



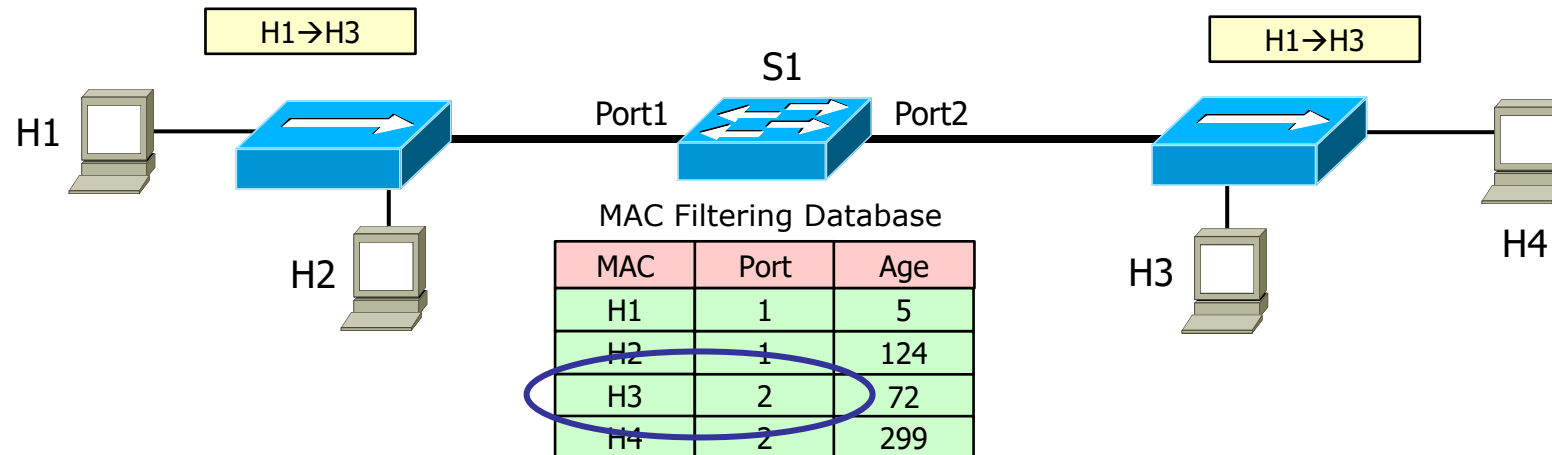
New components in “smart” bridges/switches

- In order to operate successfully, a “smart” bridge/switch requires three additional components/features:
 - A local forwarding table (*filtering database*)
 - Stations auto-learning (*backward learning*)
 - Loop detection (*spanning tree algorithm*)
- The ultimate goal: the bridge/switch should be able to do its job without any explicit configuration from the network admin
 - Really “plug and play”
- From now on we will just focus on switches



Filtering Database (1)

- Table with the “location” of any MAC address found in the network
 - MAC address
 - Destination port
 - Ageing time (default expiration after 300 s)
- “Filtering” database: in the old days, the smart forwarding process was perceived as a way to “filter out” unwanted traffic from a link



Filtering Database (2)

- Entry types
 - Dynamic
 - Populated and updated by the backward learning process
 - Max entries: 2 ÷ 64 K
 - Static
 - Not updated by the learning process
 - Usually < 1K entries
- Old dynamic entries are purged out of the filtering database
 - E.g., stations that do no longer exist on the network
 - Default: 300 seconds

Filtering database: real example

```
Cisco-switch-1> show cam dynamic
```

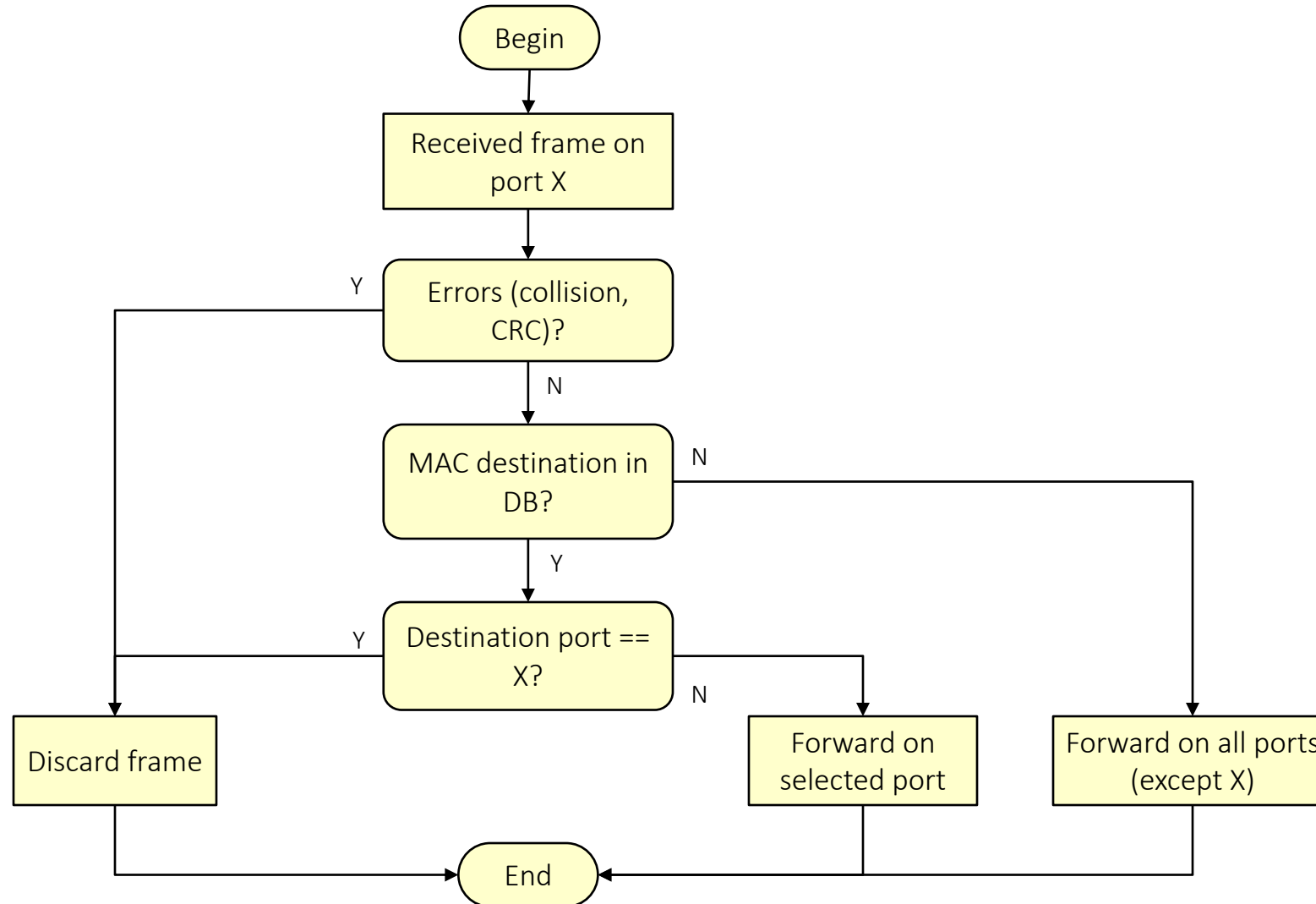
* = Static Entry. + = Permanent Entry.

= System Entry X = Port Security Entry

Dest MAC Address	Ports	Age
-----	-----	---
00-00-86-1a-a6-44	1/1	1
00-00-c9-10-b3-0f	1/1	0
00-00-f8-31-1c-3b	1/2	4
00-00-f8-31-f7-a0	1/1	2
00-01-e7-00-e3-80	2/2	0
00-02-a5-84-a7-a6	2/1	1
00-02-b3-1e-b4-aa	2/1	5
00-02-b3-1e-da-da	2/5	1
00-02-b3-1e-dc-fd	2/4	2



Forwarding process



Forwarding process and transient

- What if a MAC address is not present in the Filtering Database?
 - Switch behaves like a hub (as far as selection of ports is concerned!)
 - Frame duplicated on all ports except the one on which it was received
- This situation is rather common and it is called “transient”
 - Switches are plug-and-play and have an algorithm to learn the *location* of the hosts
 - Backward learning (presented later)
 - However, at the beginning, switches do not know where a host is located
 - In this case the “MAC Flooding” algorithm is the only way to go

How do we populate the filtering database?

- 1) By hand
 - Possible on all modern devices, but not very handy
- 2) By means of a proper algorithm
 - Backward learning
 - The best choice, of course

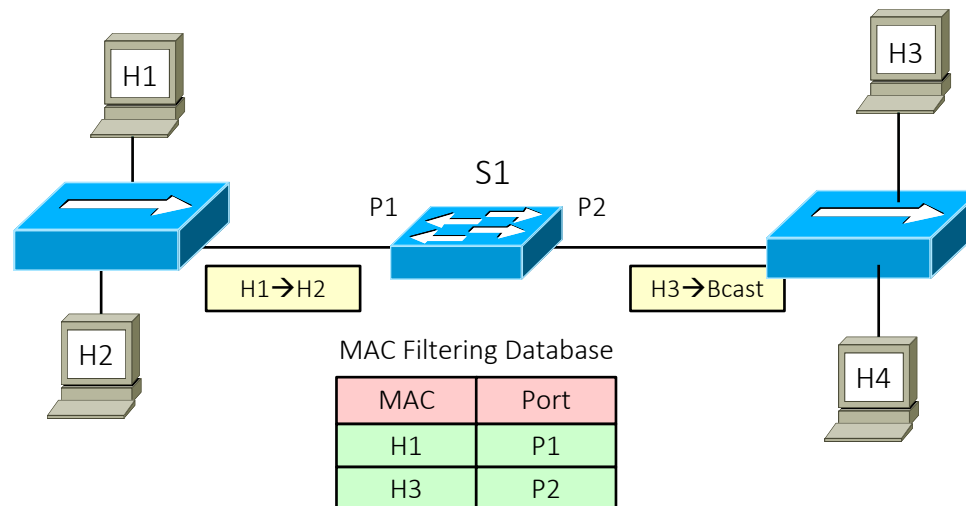
Backward learning (1)

- The idea

- If a switch receives a frame whose source is host H1 from port P1, that host will be reachable through port P1

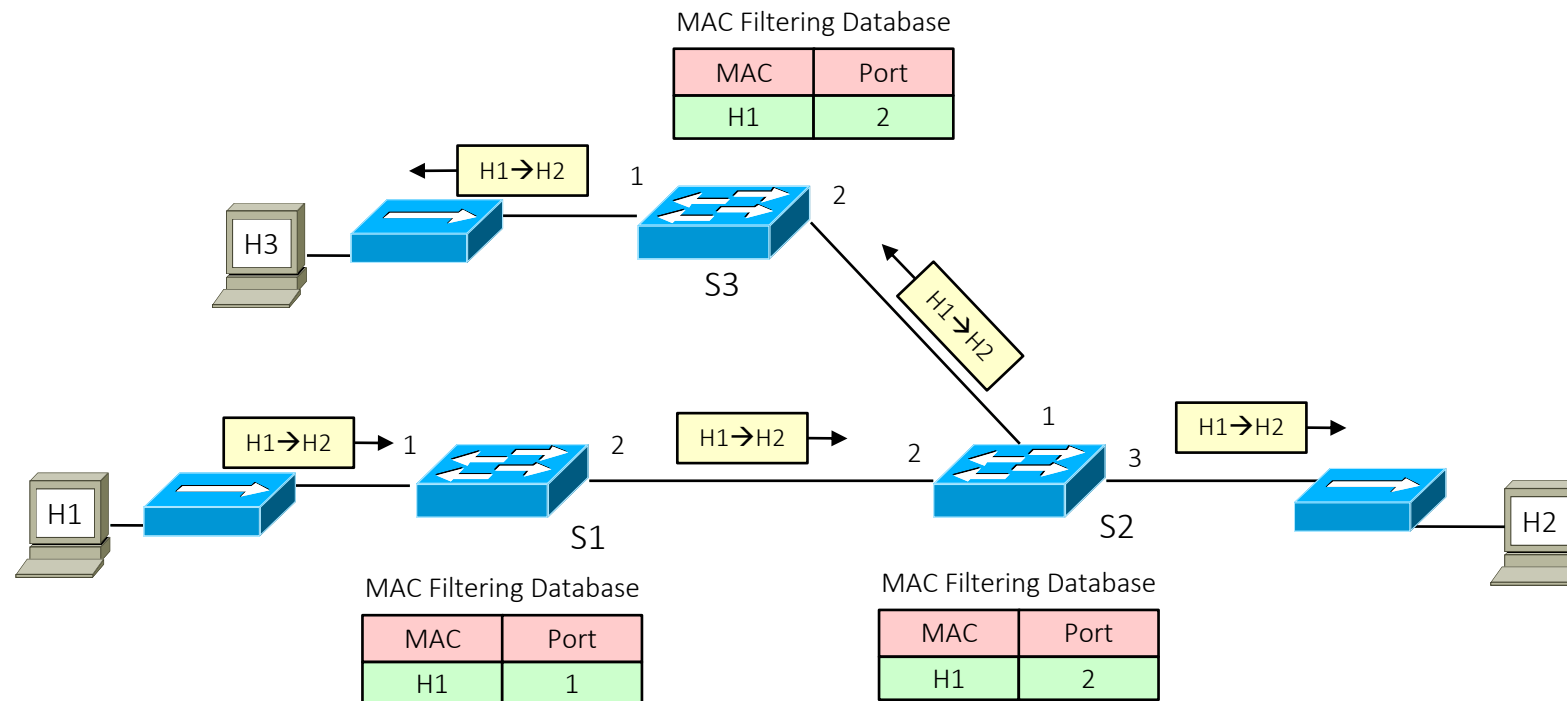
- Topology is learned by inspecting **received** frames

- Analysis of MAC source address
- The destination MAC address is ignored by this algorithm

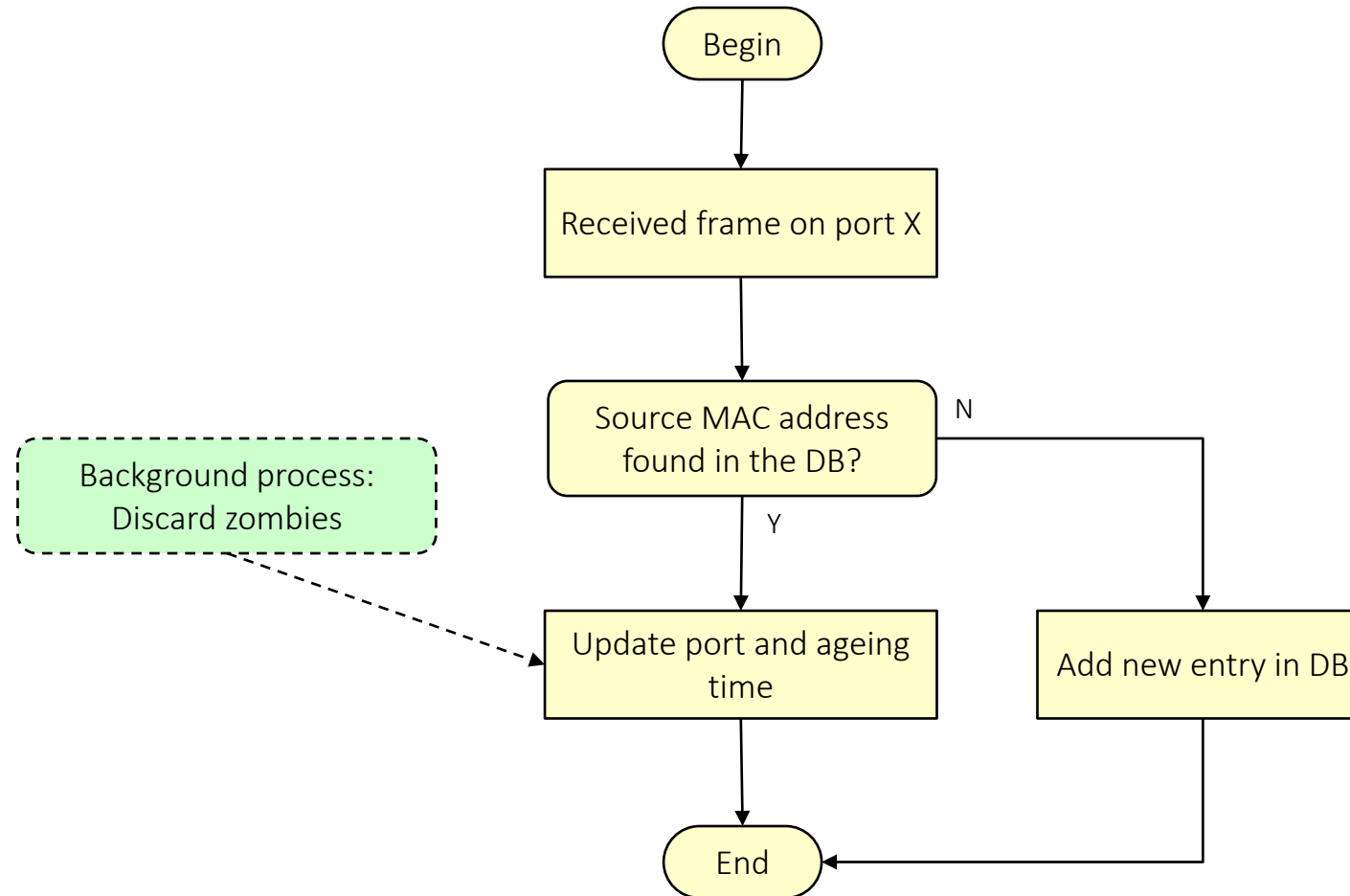


Backward learning (2)

- Works also in presence of multiple switches
 - Remote switches learn the position anyway, even if end-system not connected locally
- Example: backward learning and frame forwarding taken together



Backward learning (3)

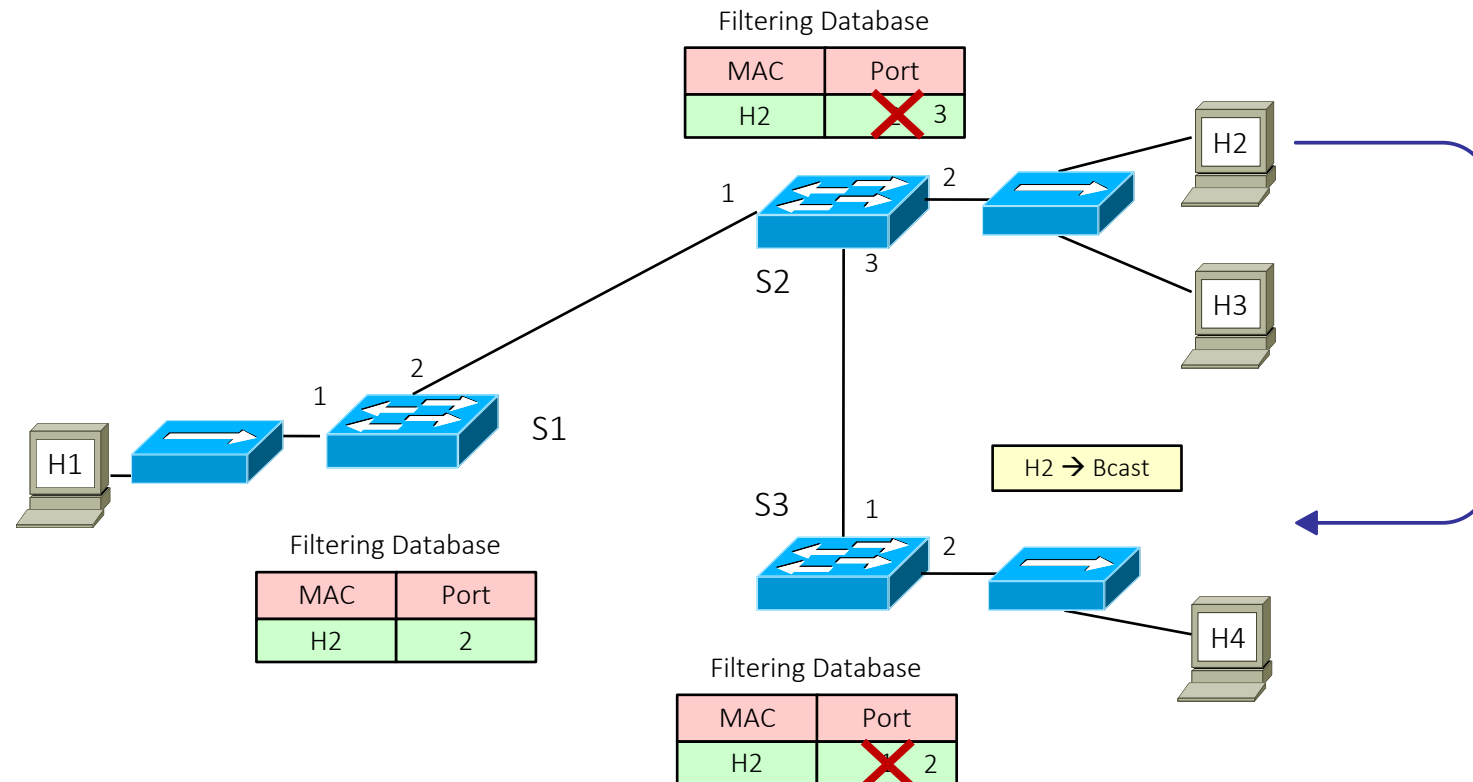


How do we keep the filtering DB up to date?

- Update the Filtering database means...
 - Refresh “Age”, so that the entry keeps alive
 - Refresh “Port”, so that the host is updated with the new position
- Note that...
 - An end-system whose MAC address is not in the DB is *always reachable*
 - Corollary: a frame sent to a non-existing host will always be forwarded in all the network
 - An end-system whose MAC address is in the DB may be *unreachable*
 - At most for *Aging Time*, in fact

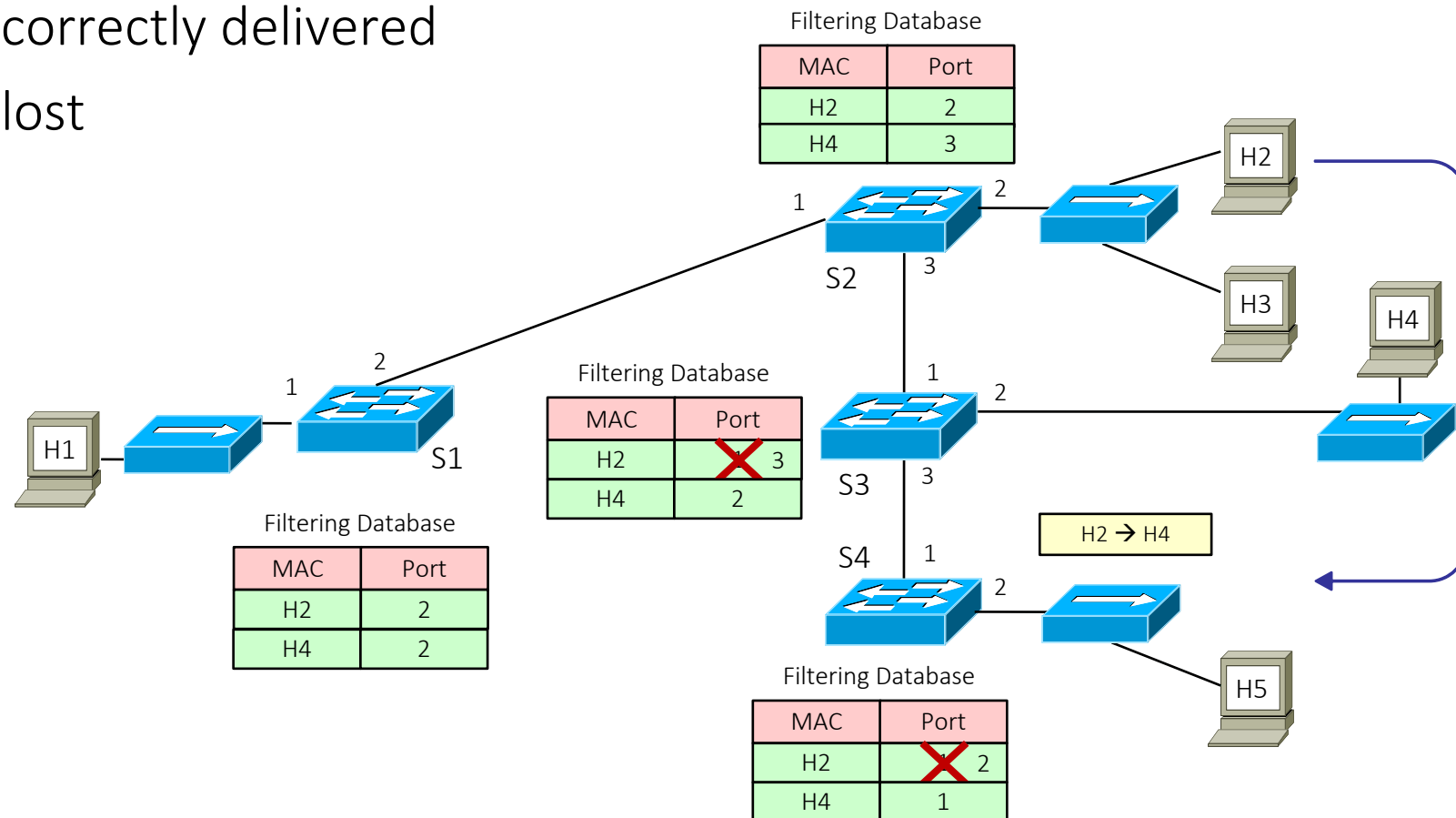
L2 networks and hosts mobility (1)

- If the end-system generates broadcast frame immediately
 - No problems



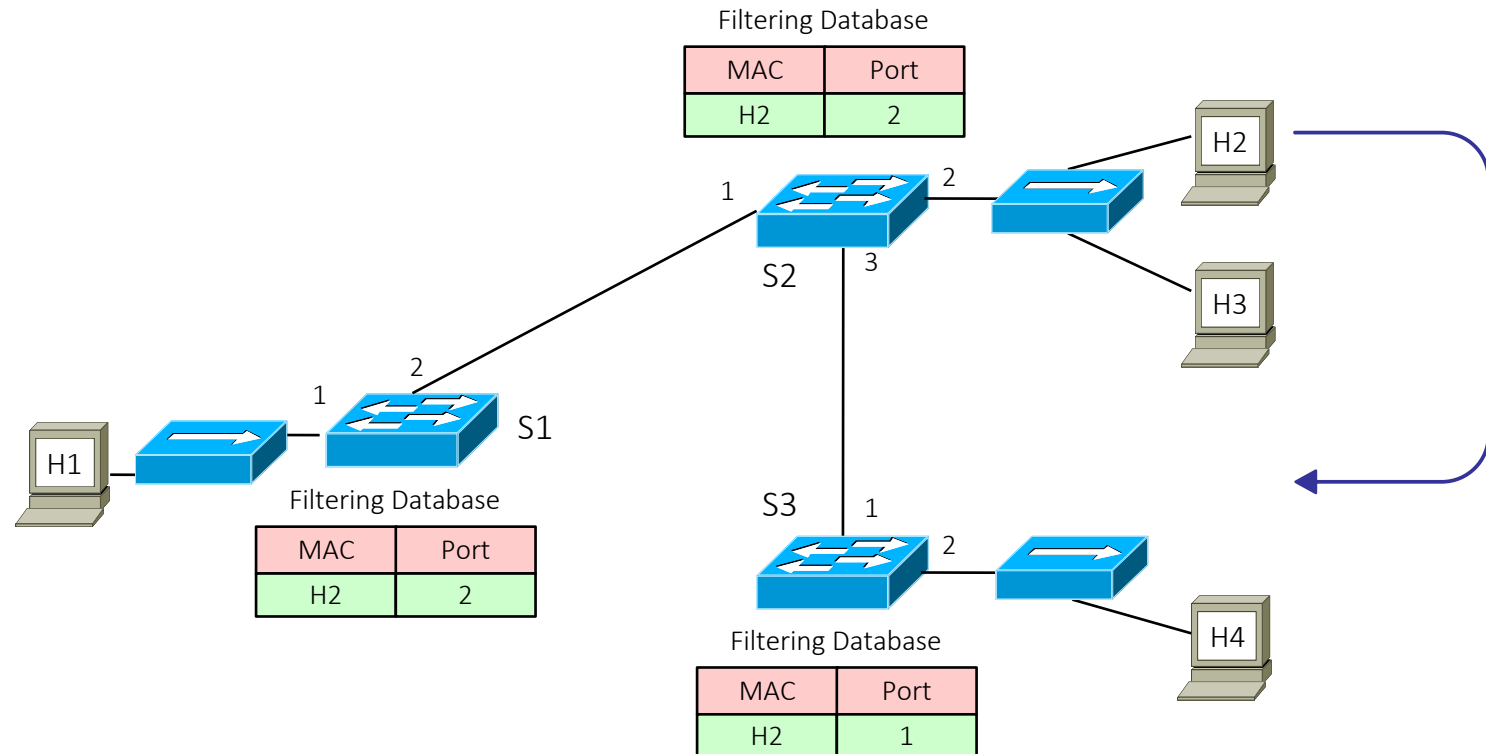
L2 networks and hosts mobility (2)

- If the end-system generates unicast traffic immediately
 - We may have forwarding errors
 - H4 → H2 is correctly delivered
 - H3 → H2 is lost



L2 networks and hosts mobility (3)

- If the end-system does not generate traffic at all
 - We may have forwarding troubles
 - H4 → H2 is correctly delivered
 - The frame is forwarded also to the original destination
 - H3 → H2 is lost



L2 networks and hosts mobility (4)

- Broadcast (multicast) frame
 - Reaches entire network, therefore all the switches update location of current station
- Unicast frame
 - Potentially reaches only a portion of the network, hence the rest may still have the old location of the station
- In the real world
 - Usually enough in order to cope with manual movements
 - A laptop moved from office to lab
 - Possible problems in specific environments
 - E.g., fault-tolerant NICs
 - We need to react much quickly than 5min
 - NIC driver might have to generate an additional broadcast frame



Possible attacks to the filtering database

■ MAC Flooding Attack

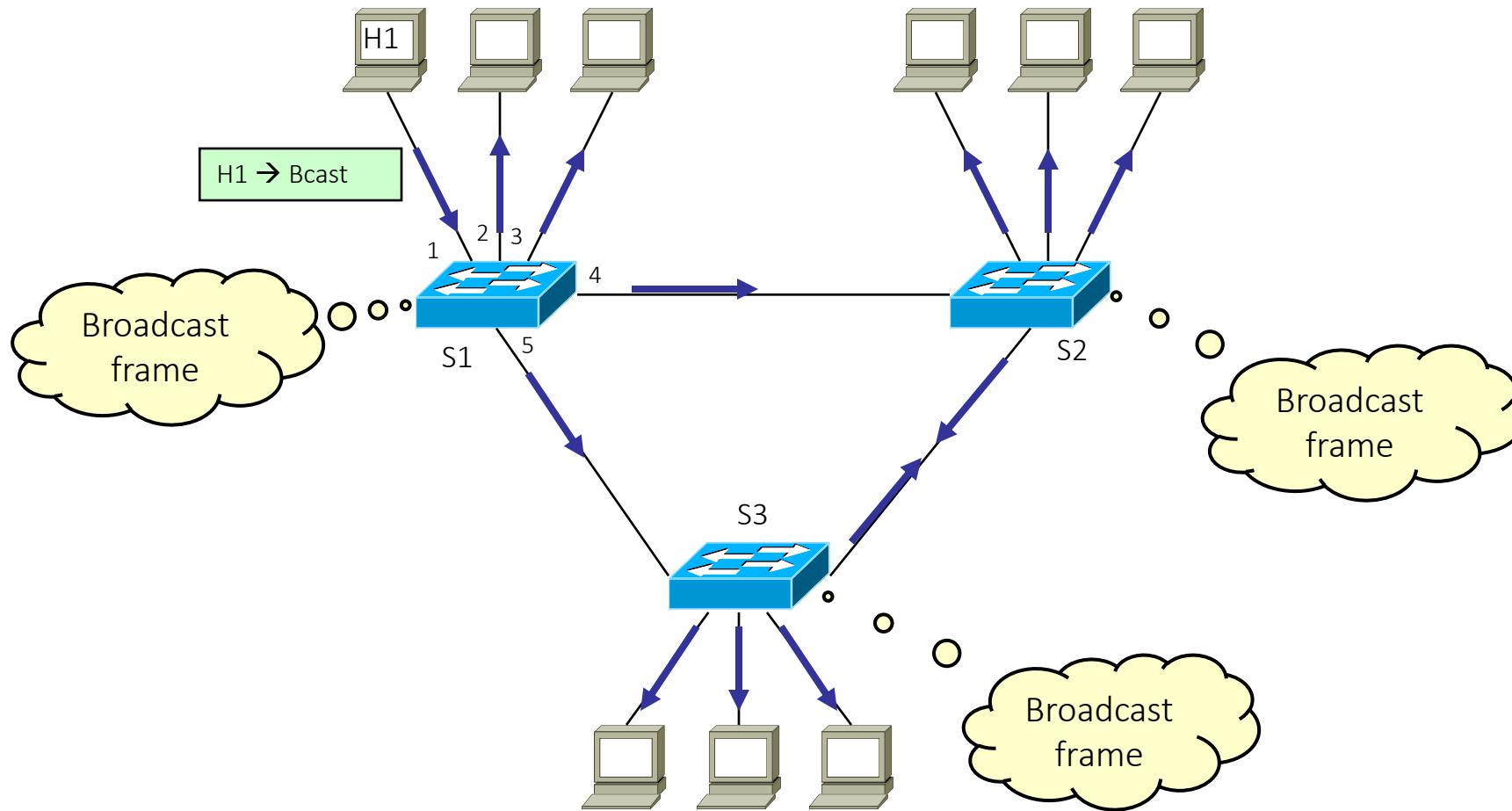
- Generation of frames with random MAC sources
- Filtering database gets full
- Switches will start flooding most of the frames
 - All the ones whose destination address is not present in the DB
- Objectives
 - Forces switches to operate like hubs, so that we can intercept traffic generated by other stations
 - Slows down the network
- Some vendors give the opportunity to limit the number of MAC addresses learnt on each port



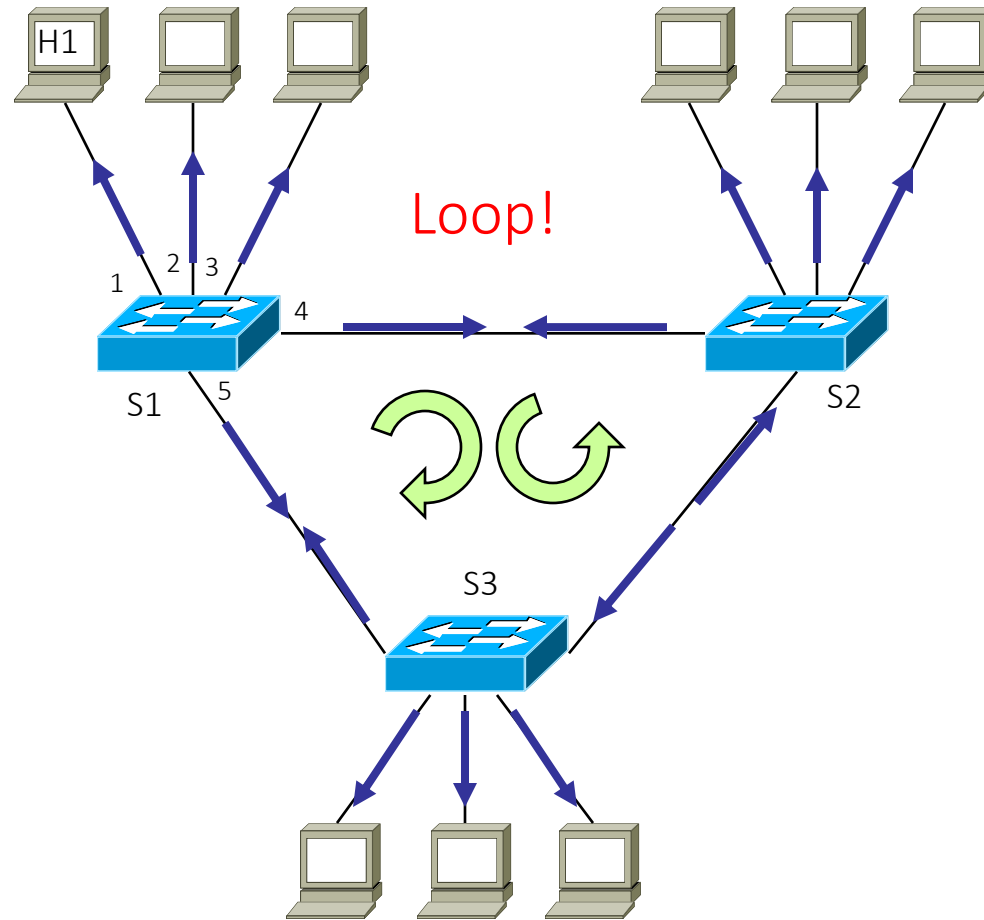
Switches and meshes

- Two problems
 - Frames can enter in a loop
 - Backward learning no longer able to operate
- It's now time to introduce the third component:
 - A local forwarding table (*filtering database*)
 - Stations auto-learning (*backward learning*)
 - Loop detection (*spanning tree algorithm*)

Switches and meshes: the loop problem



Switches and meshes: the loop problem



Which frames can generate a loop?

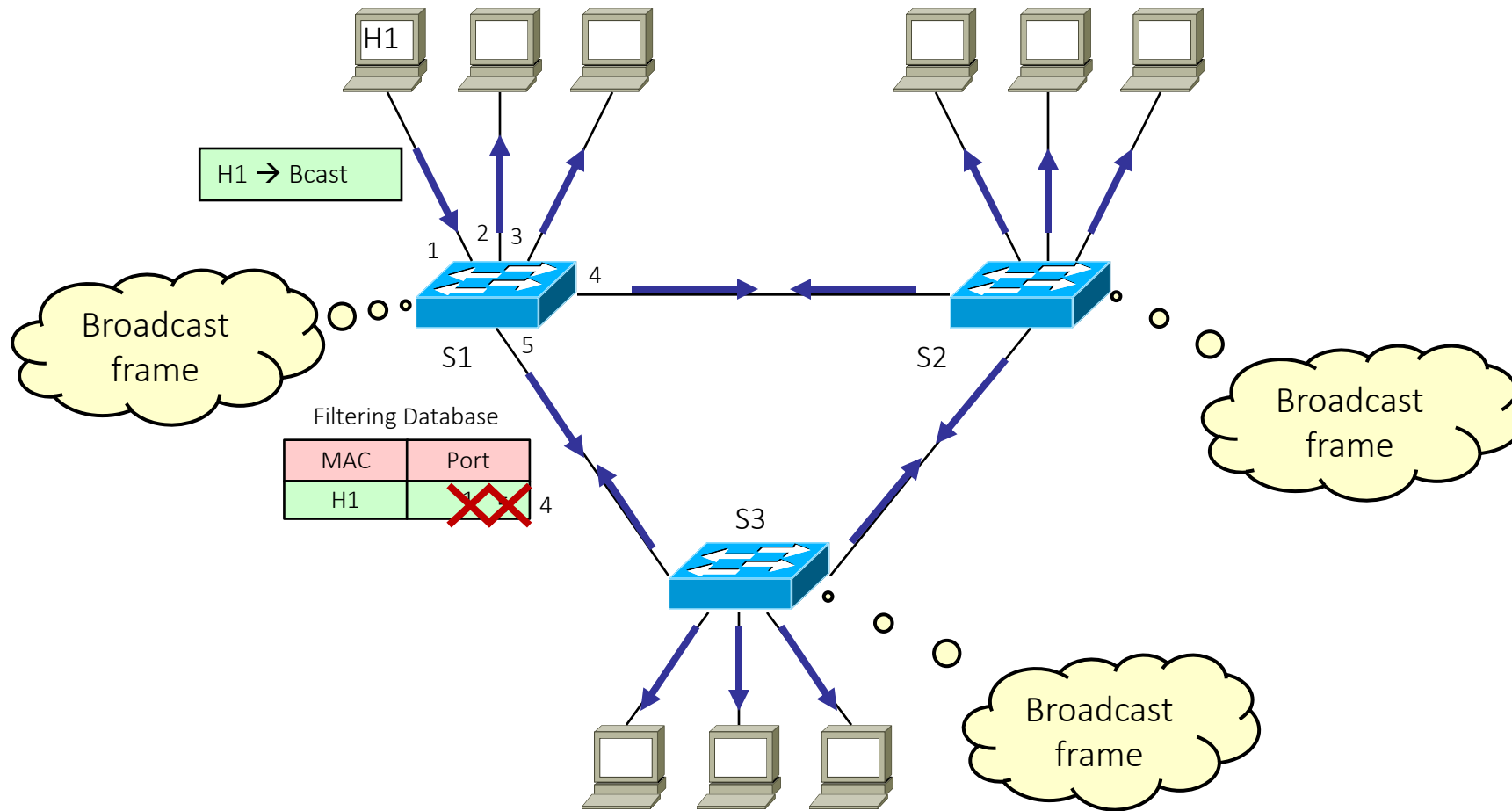
- Multicast/broadcast frames
 - Very common
- Frame to a non-existing station
 - MAC address not present in the filtering DB (e.g. non existing station)
 - Problem that may happen rarely (unless under attack)
 - IP sends an ARP before contacting an L2 station
 - If the station does not exist, the ARP will never get a reply and the destination MAC address is unknown
 - Therefore, no MAC frames will be sent to that station intentionally

The Broadcast Storm

- Massive load due to broadcast/multicast traffic on a LAN
- One of the most dangerous problems at data-link layer
- No solutions, except for disabling (physically) loops
 - E.g., detach a cable from a bridge
- Network operators are almost impotent in such a case
- Due to the lack of a “time-to-live” field in L2 frames
- L3 networks can tolerate transient loops
 - TTL available on L3 packets
- Can be used to create a low-cost traffic generator sending frames at line-rate



Switches and meshes: the learning problem (1)

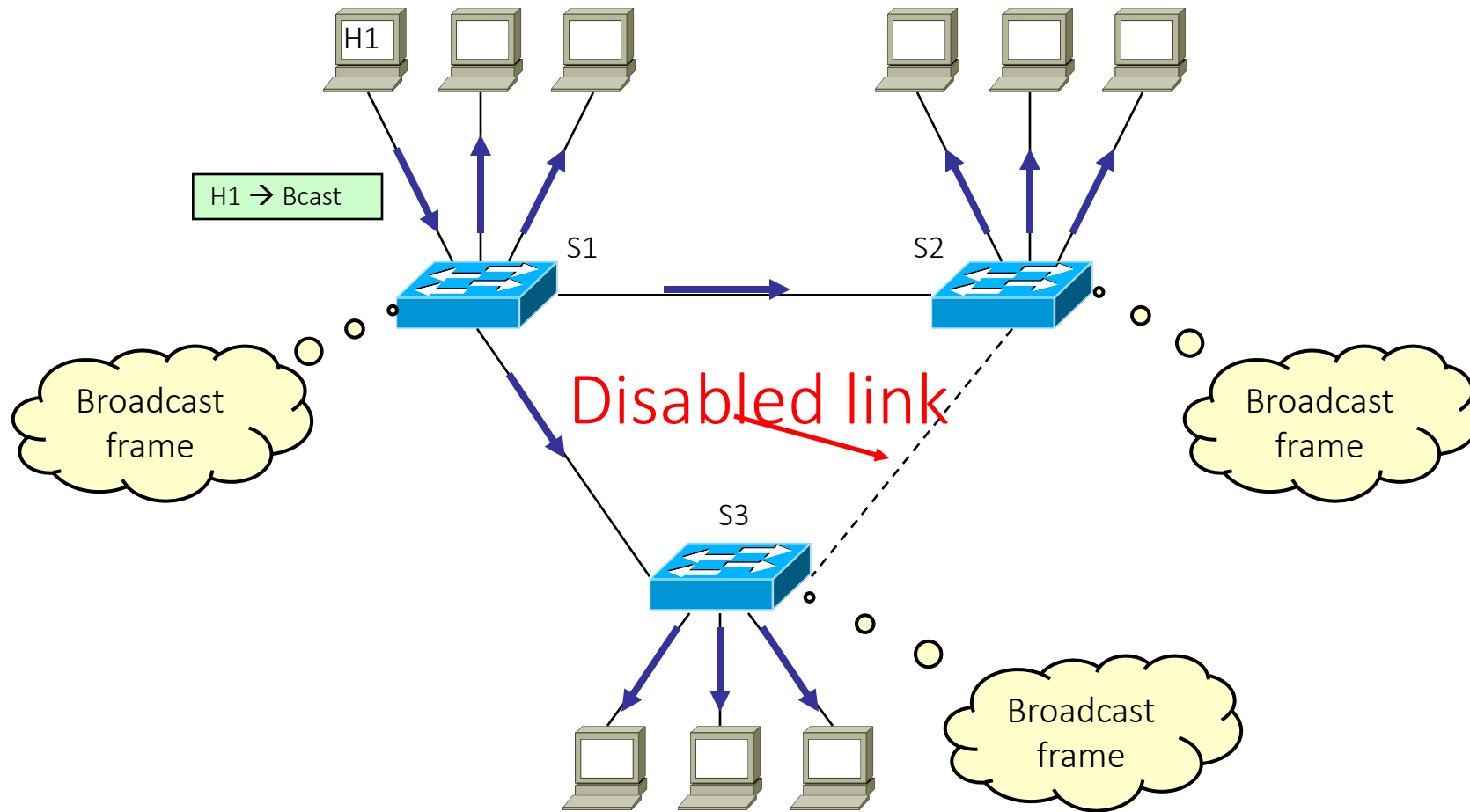


Switches and meshes: the learning problem (2)

- Backward learning problem
 - Switches may have inconsistent filtering database
 - An entry in the filtering database may change the port indefinitely
 - An entry may not be able to reach a stable state
 - Transient loops can be created among back-to-back switches
 - B1 forwards to B2 that forwards to B1,...
 - Larger (B1-B2-B3-B1) loops may occur as well



The Spanning Tree idea: no loops in the network



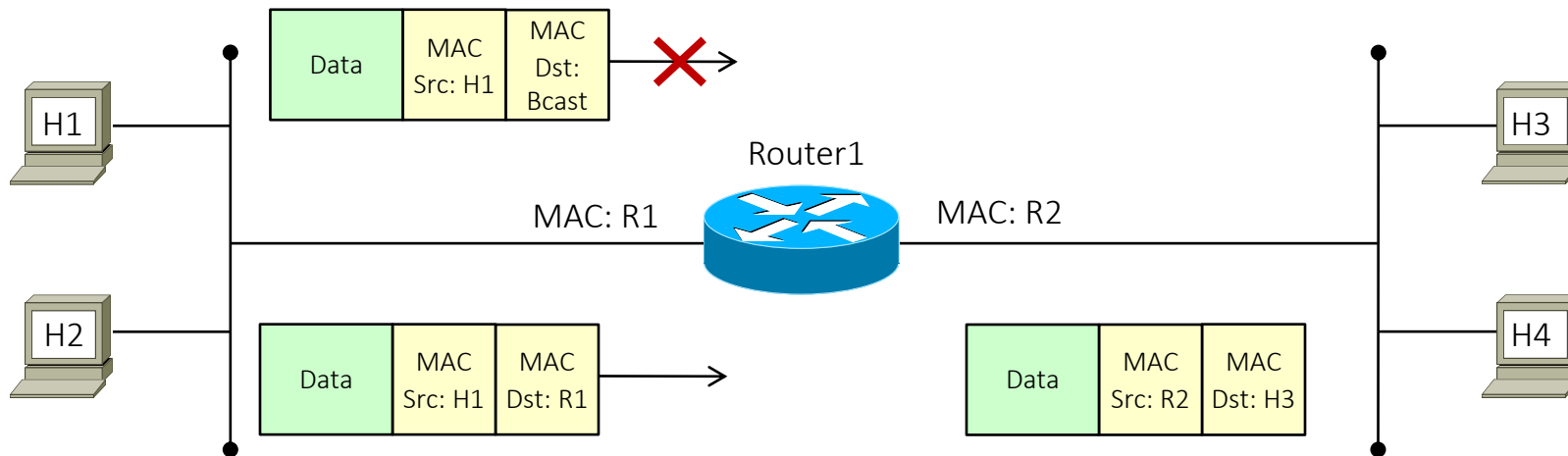
Spanning Tree

- In order to avoid troubles, you must avoid loops in the physical network
 - Either create loop-free networks
 - Discouraged; not robust
 - Or define an algorithm that disables (temporarily) loops
- 802.1D
 - Original idea from Radia Perlman, PhD @DEC
- Meshes detected and disabled; the network becomes a tree
 - Unique path between any source and any destination

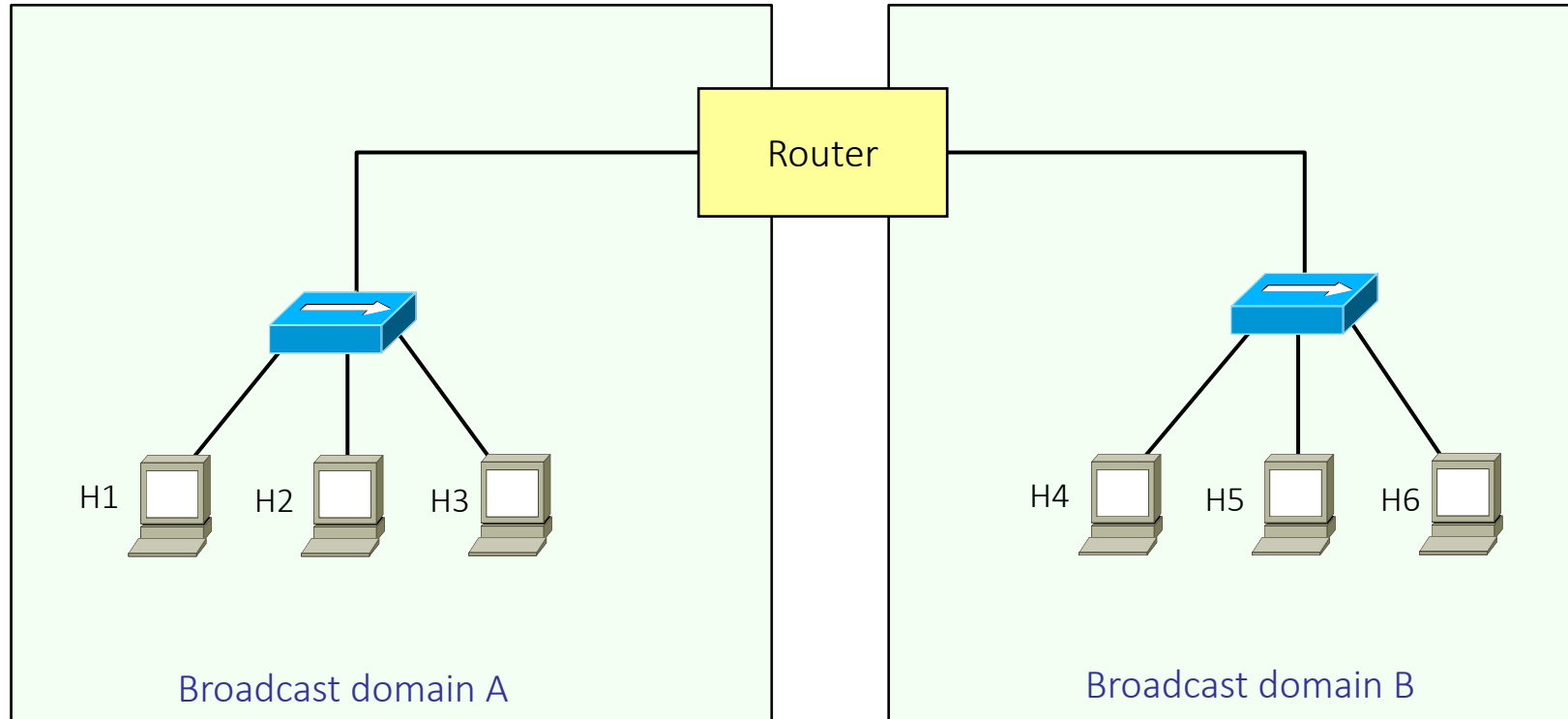


Routers

- L3 devices!
 - Routers are not transparent with respect to MAC addresses
 - Routers separate **broadcast domains**



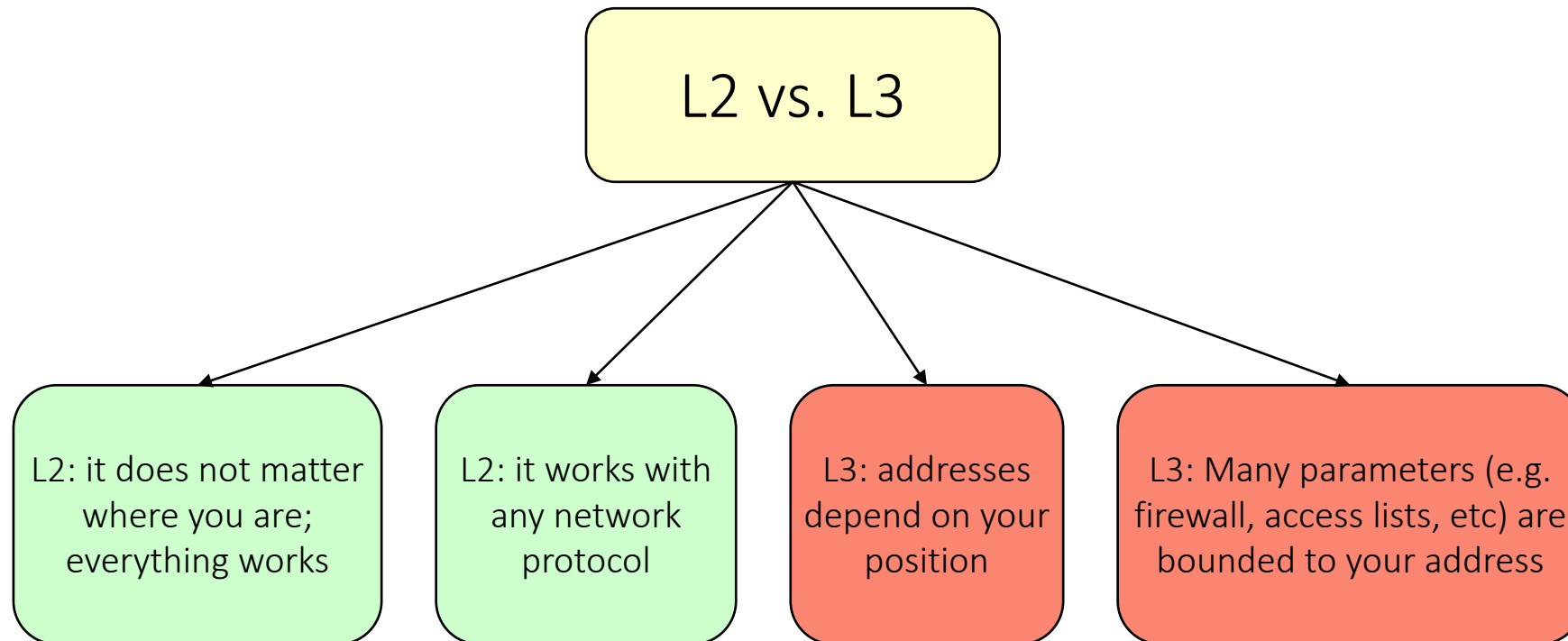
Routers and broadcast domains



Different IP networks on the two interfaces of the router

L2 or L3?

- So far, we concentrated on L2
- Shall we stay with L2 or better moving to L3?

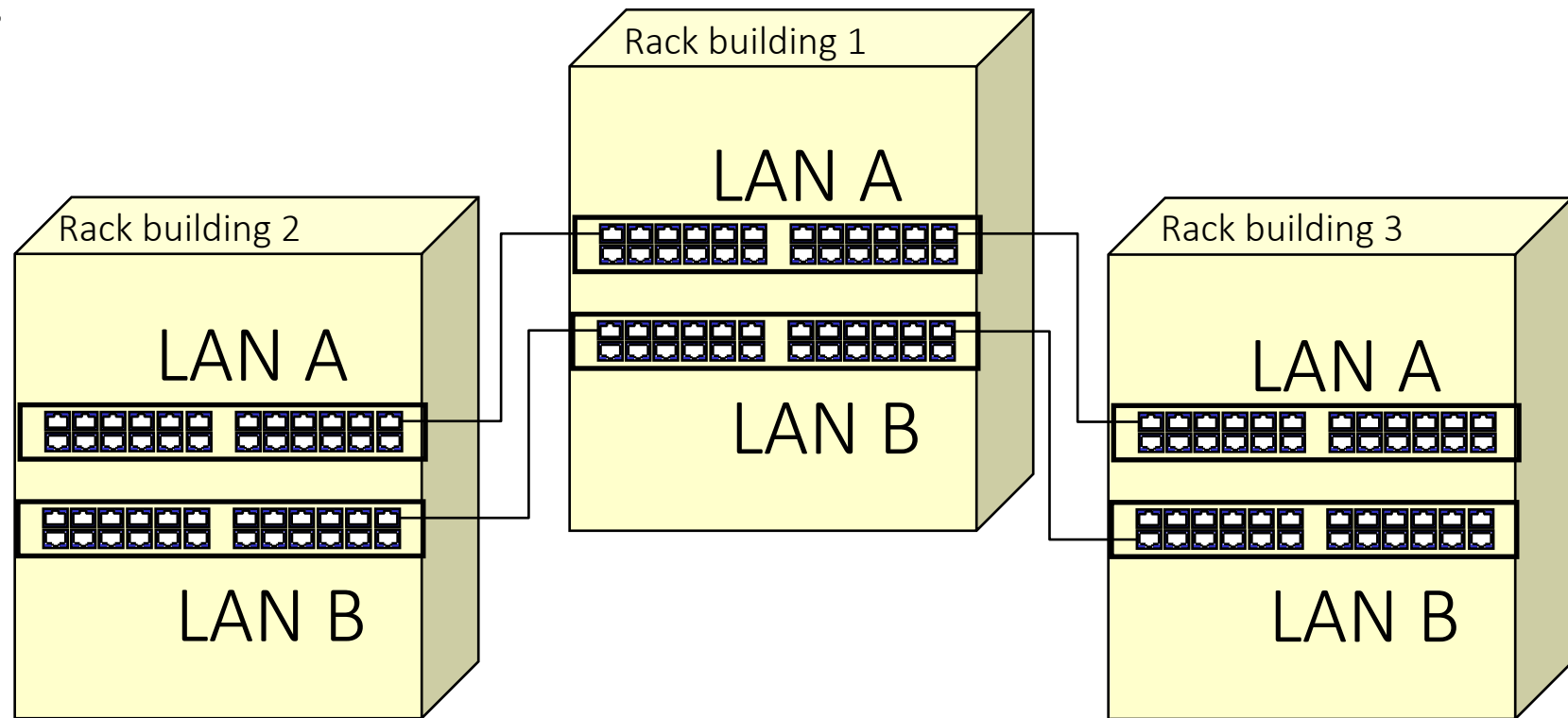


One or multiple LANs across a campus?

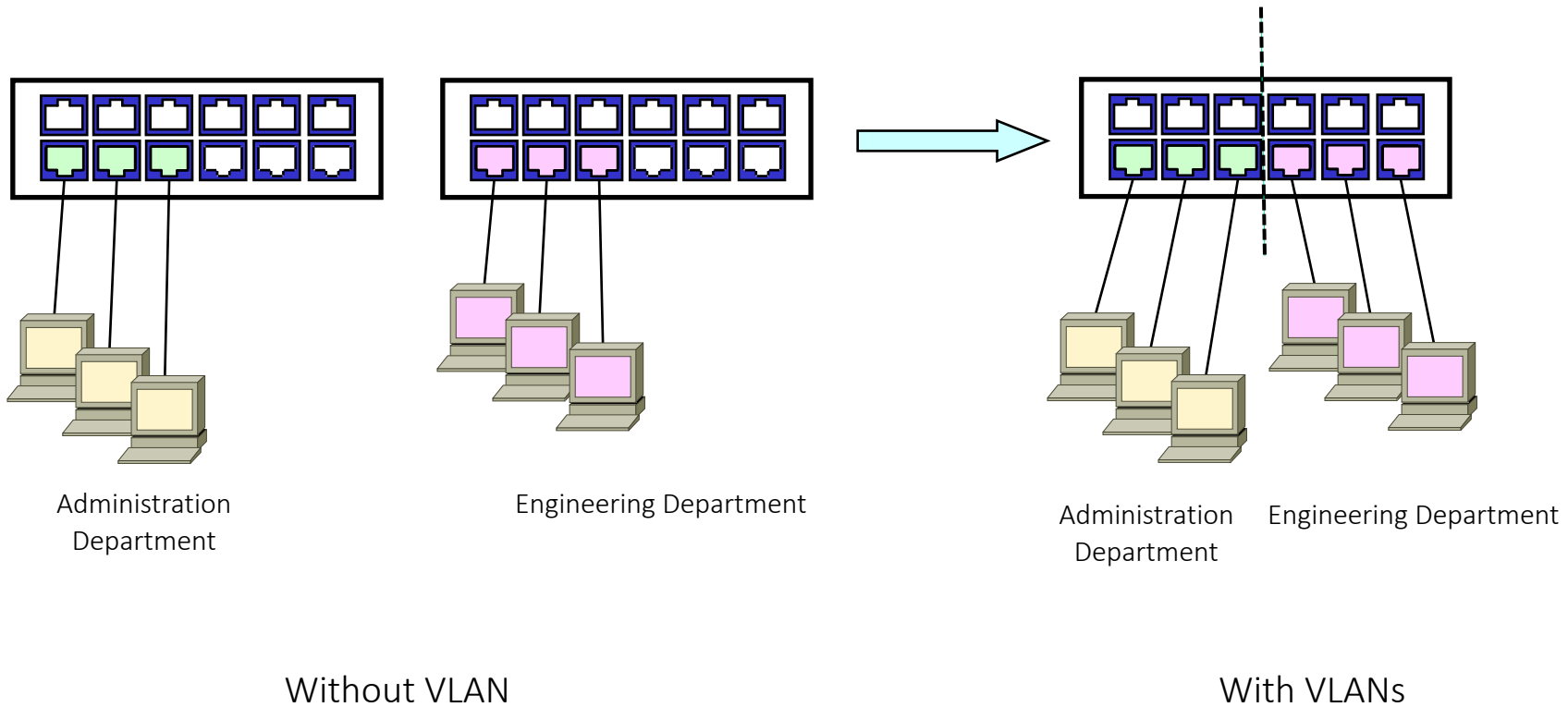
- Ok, so it's better to keep the L2 as long as we can
 - As long as the network can operate as a single L2 entity (remember scalability issues in L2 networks!)
- But... a single, gigantic LAN, or multiple LANs?
 - Performance
 - A single LAN has too much broadcast traffic (not filtered by switches)
 - Flooded traffic (e.g., due to frequent STP reconfiguration)
 - Privacy, Security
 - Do not want a station to leak some information out (e.g., MAC Flooding attack)
 - Management
 - Smaller network, simple (and uniform) policies
- Better to partition different users in different LANs

Multiple LANs across a campus: how?

- Different physical networks (full separation)
 - N networks = N links + N devices
 - Waste of resources



Virtual LANs (1)

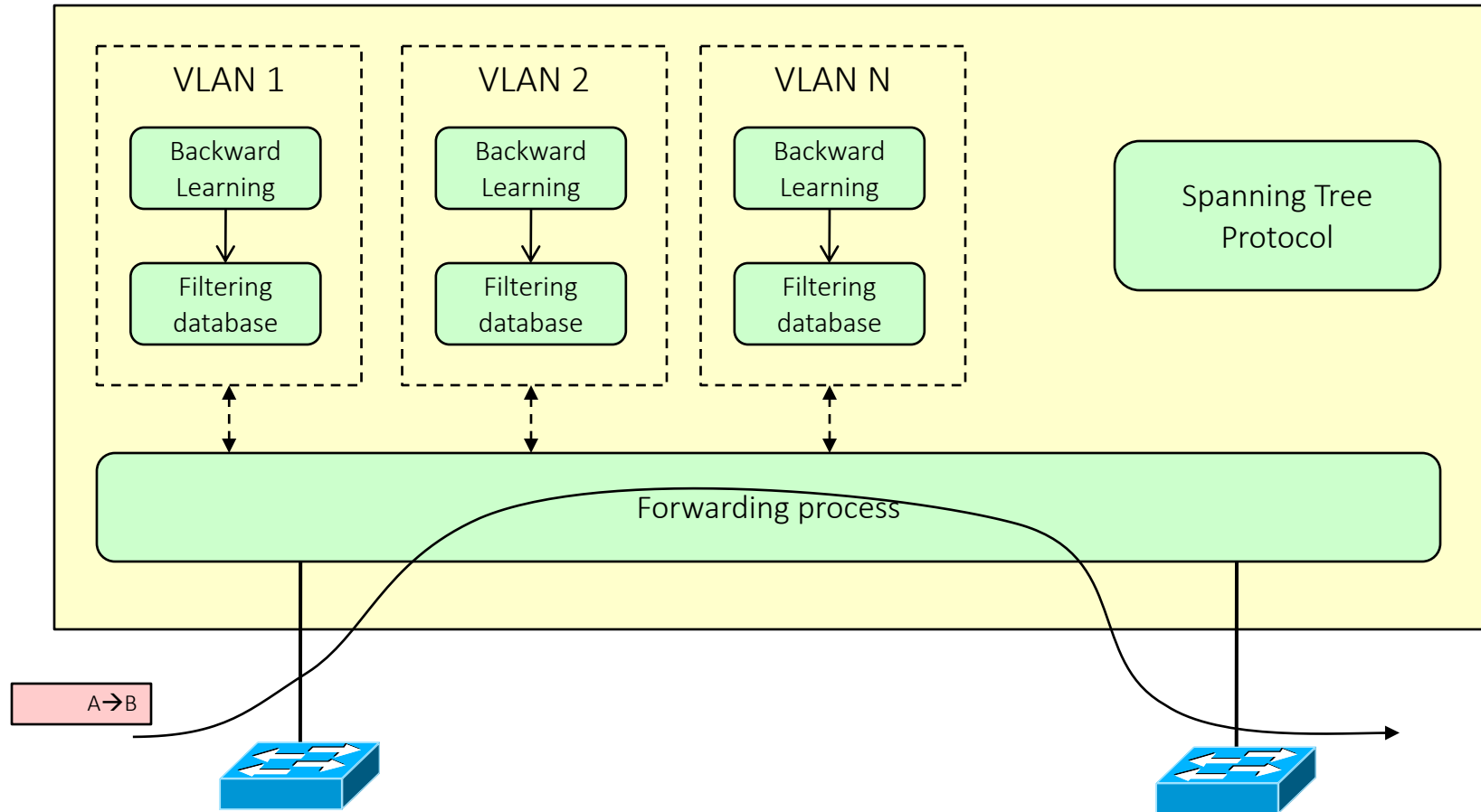


Virtual LANs (2)

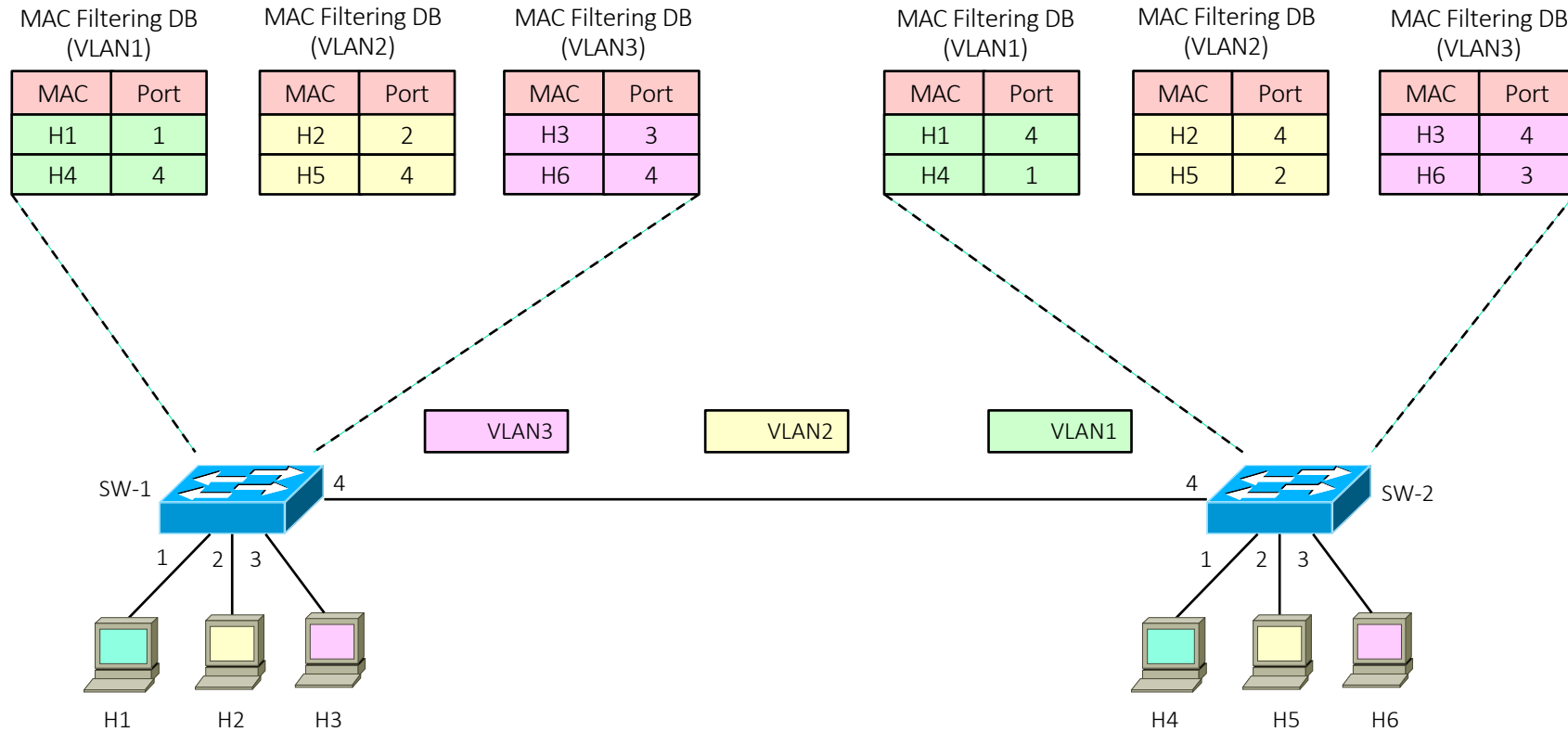
- Single physical infrastructure
 - Same devices, same cabling
 - No switches in which only a few ports are used
 - No need to have multiple fibers (for different LANs) in the backbone
- Different LANs
 - Different broadcast domains
 - E.g., Ethernet frames cannot be propagated on another VLAN
 - No broadcast between LANs
 - No MAC flooding attacks
 - No ARP spoofing
 - Created through a proper (logic) separation on switches
 - Intra-switch or inter-switch



VLAN: switch architecture



VLAN: forwarding database



Real implementations: unique filtering database (usually made with a TCAM, which is a single entity in the network device)

Interconnecting VLANs (1)

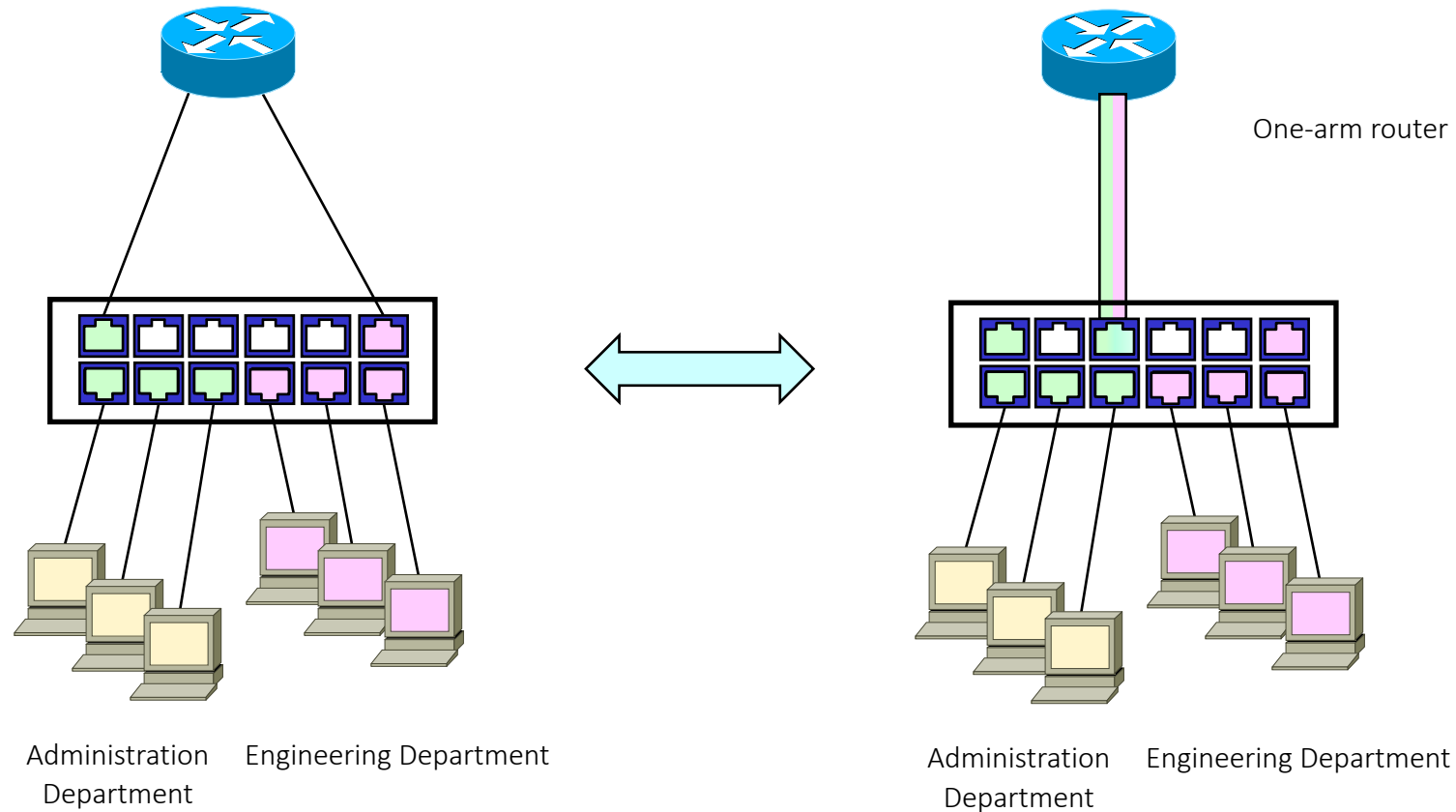
- L2 data cannot cross VLANs
 - An Ethernet station cannot send an L2 frame to another station in a different VLAN
 - VLANs are different broadcast domains

Beware:

L2 data cannot cross VLANs!



Interconnecting VLANs (2)

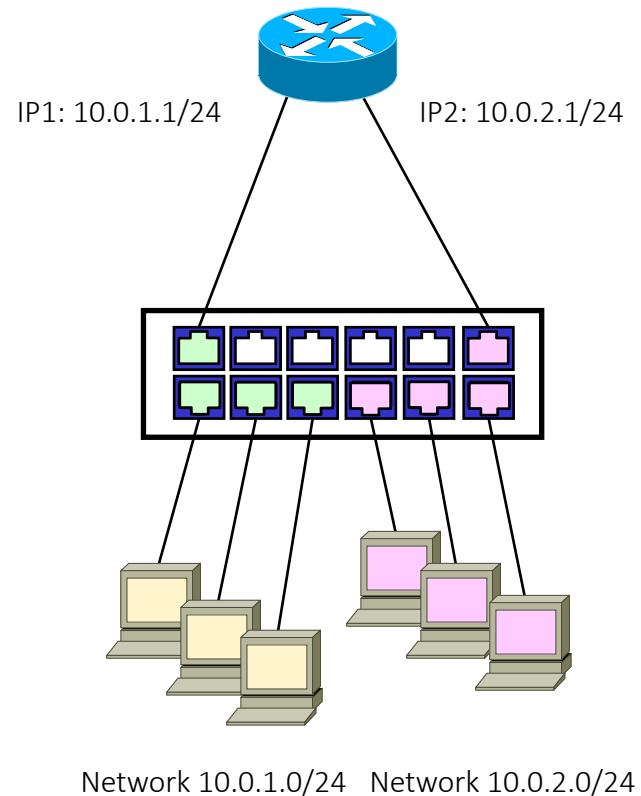


Interconnecting VLANs (3)

- A router (i.e., device operating at layer 3) is needed
 - Lookup at layer 3 (e.g., IP destination address)
 - The original L2 header is thrown away and a new one is created with other MAC addresses (src/dst)

VLANs and IP addresses

- Broadcast cannot cross the VLAN boundaries
 - Cannot use ARP to resolve the MAC address in another VLAN
- Hosts in different VLANs must belong to different IP networks



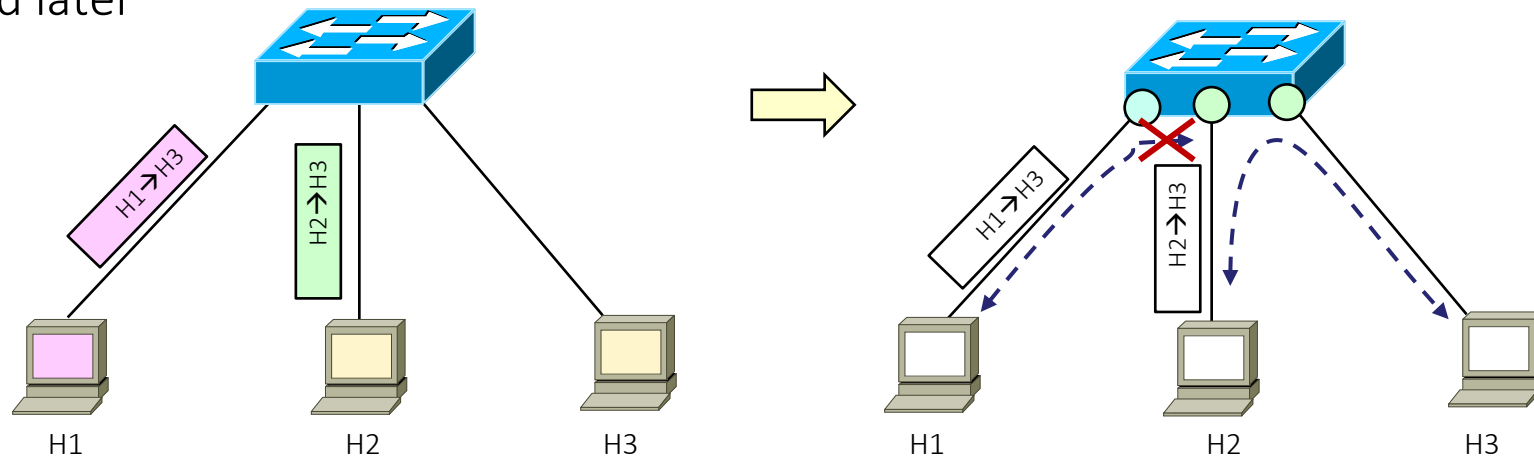
Associate frames to VLANs (1)

- Problem

- How can we associate frames to VLANs?

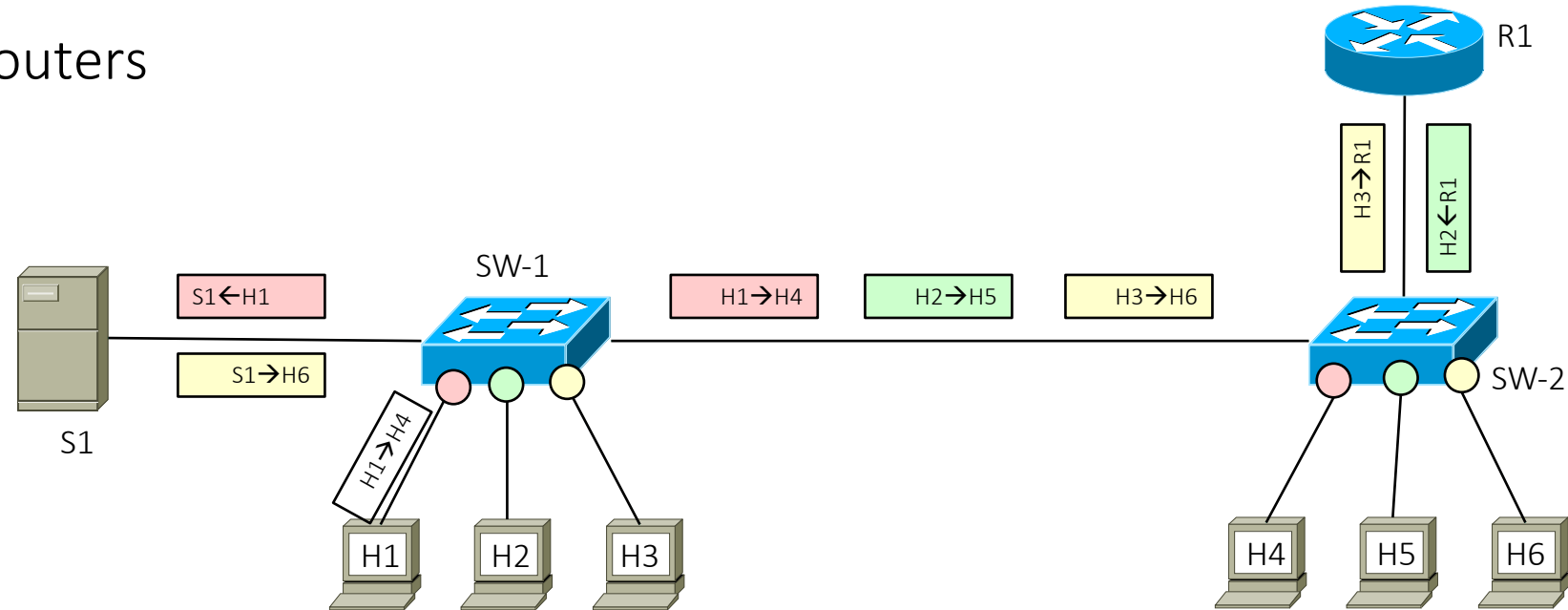
- VLANs on a single switch

- Simplest method: we can mark the ports on the switch
 - The received frame is associated to the VLAN the port belongs to
 - Other methods exist
 - Briefly presented later



Associate frames to VLANs (2)

- VLAN on different switches
 - Problem: how to distinguish which VLAN a frame belongs to, as there is a single link between switches?
- Same problem for devices that belong to different VLANs
 - E.g., servers, routers

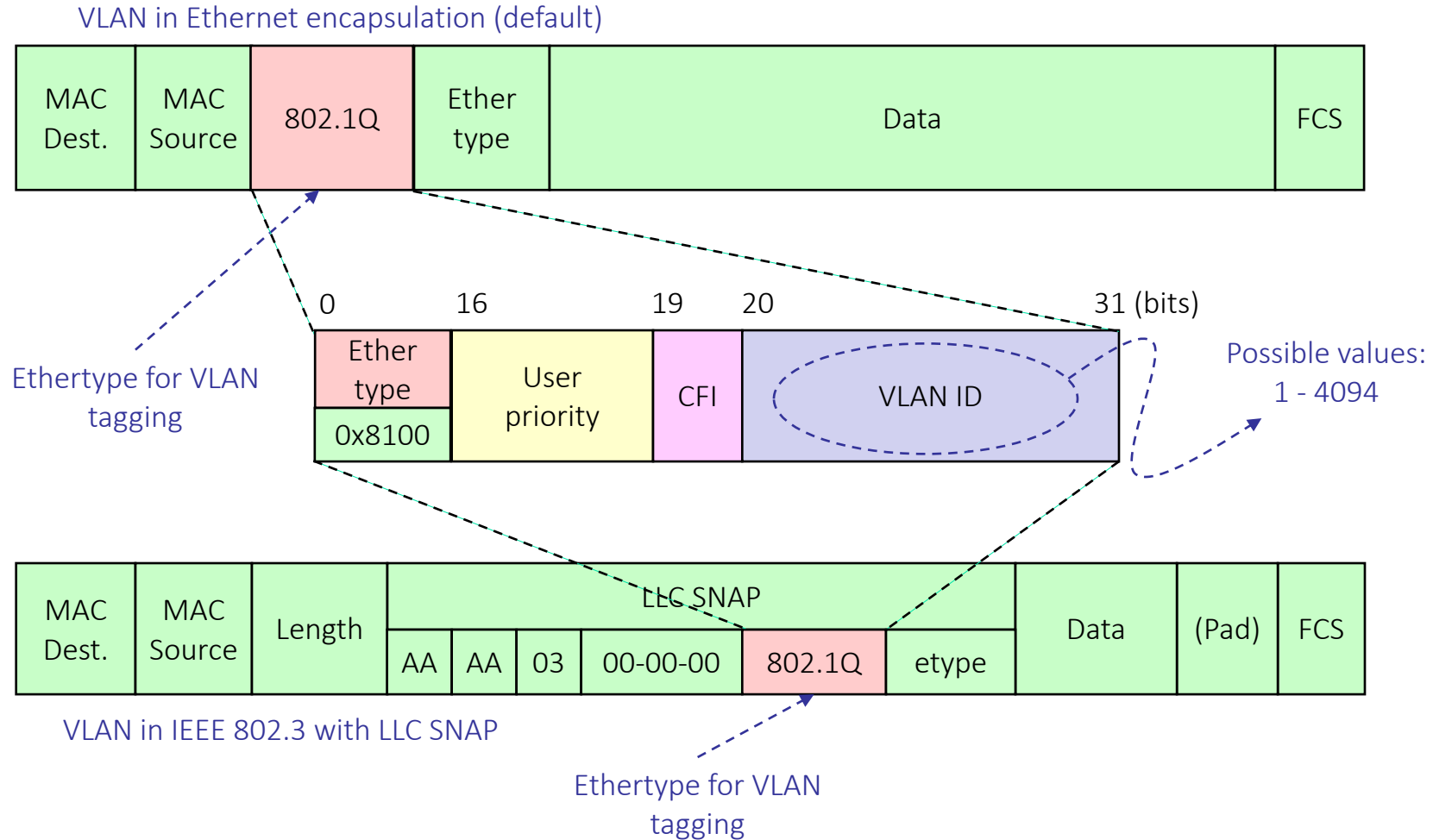


Note: the IDs in the frames are the MAC addresses of the involved stations

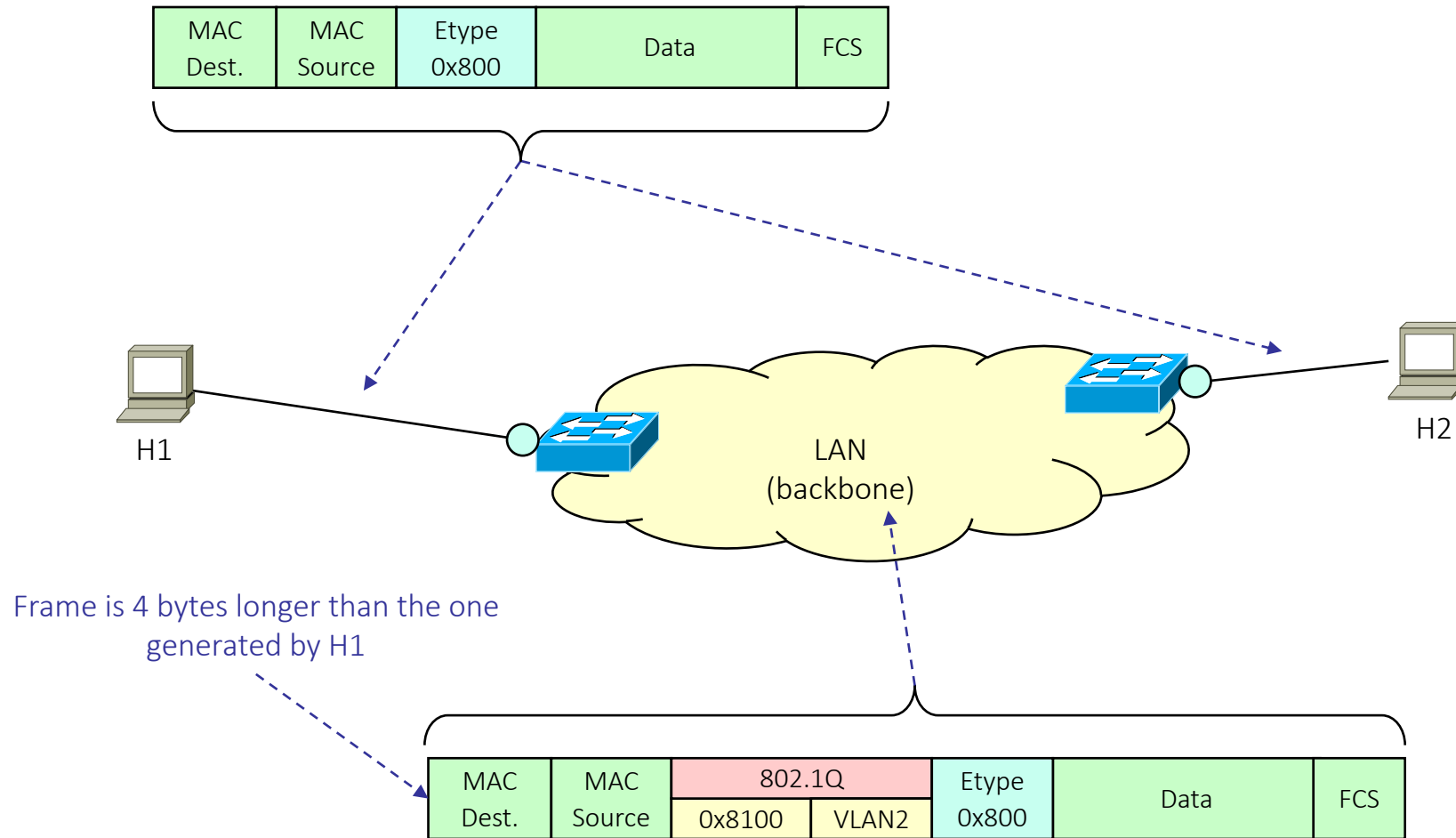
Associate frames to VLANs: tagging

- Required only on links that transport traffic of different VLANs
- An additional header is added to the MAC header
 - Standardized by IEEE 802.1Q
 - 4 additional bytes added to the frame
 - Basically, VLAN-ID plus a bunch of other info

IEEE 802.1Q Tag Encoding (1)



IEEE 802.1Q Tag Encoding (2)

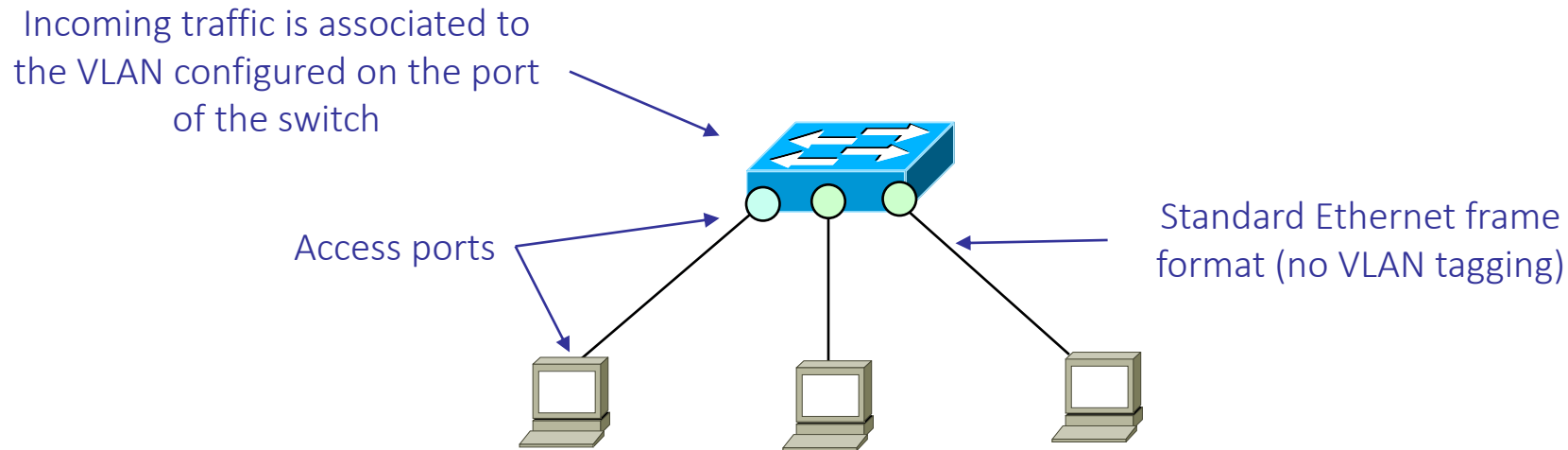


Modification to existing MACs

- Minor modifications
- New framing (for tagging) specified in 802.1Q
 - Independent from the technology of the Medium Access Control
- Maximum length of the frame has to be extended 4 bytes
 - E.g., Ethernet reaches 1522 bytes (from 1518)
 - Minimum length unchanged (still 64 bytes)

Port types: Access (1)

- *Access Ports* receive and transmit *Untagged* frames
- Default configuration (on hosts, switches, servers, routers, etc)
- Usually used to connect end-stations to the network
 - Hosts do not need to change their frame format



Port types: Access (2)

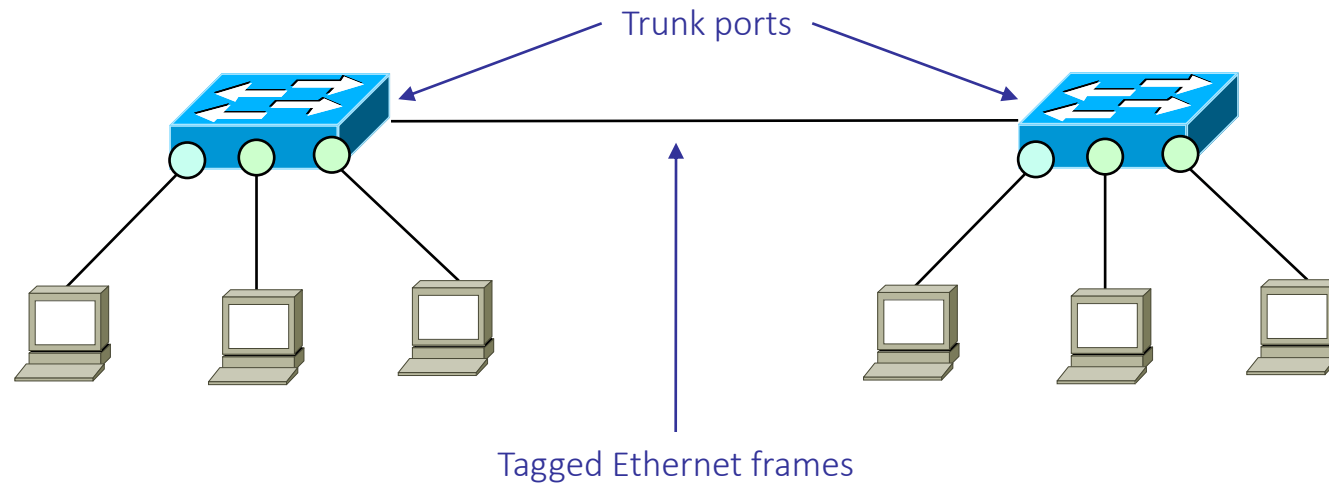
- Given the following network
 - All ports are configured in “access mode”
 - SW-1 is configured with the RED VLAN on all its ports
 - SW-2 is configured with the GREEN VLAN on all its ports
- Can host H1 communicate with host H4?



Yes, because values configured on access ports are not propagated outside the switch!

Port types: Trunk

- Trunk ports receive and transmit Tagged frames
- Must be configured explicitly
 - Often used in switch-to-switch connections and to connect servers/routers



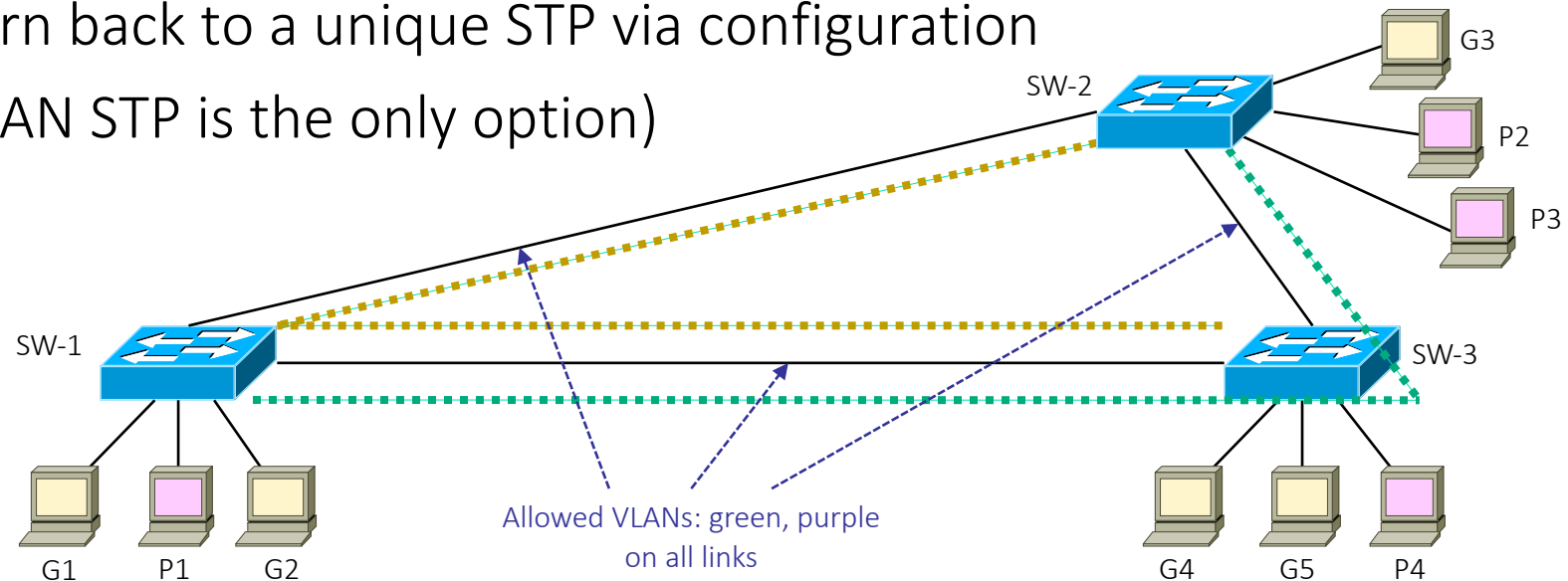
Assigning hosts to VLANs

- Different methods to associate devices to the proper VLAN
 - Port-based VLANs
 - Transparent assignment
 - E.g., Associated to MAC addresses
 - Per-user assignment (802.1x)
 - Cooperative assignment
 - Frames tagged directly by hosts
- Note: a station can be associated also to multiple VLANs
 - E.g., required in case of servers, routers
 - In this case, trunk ports are required on the device
 - Frames tagged by the device (cooperative assignment)



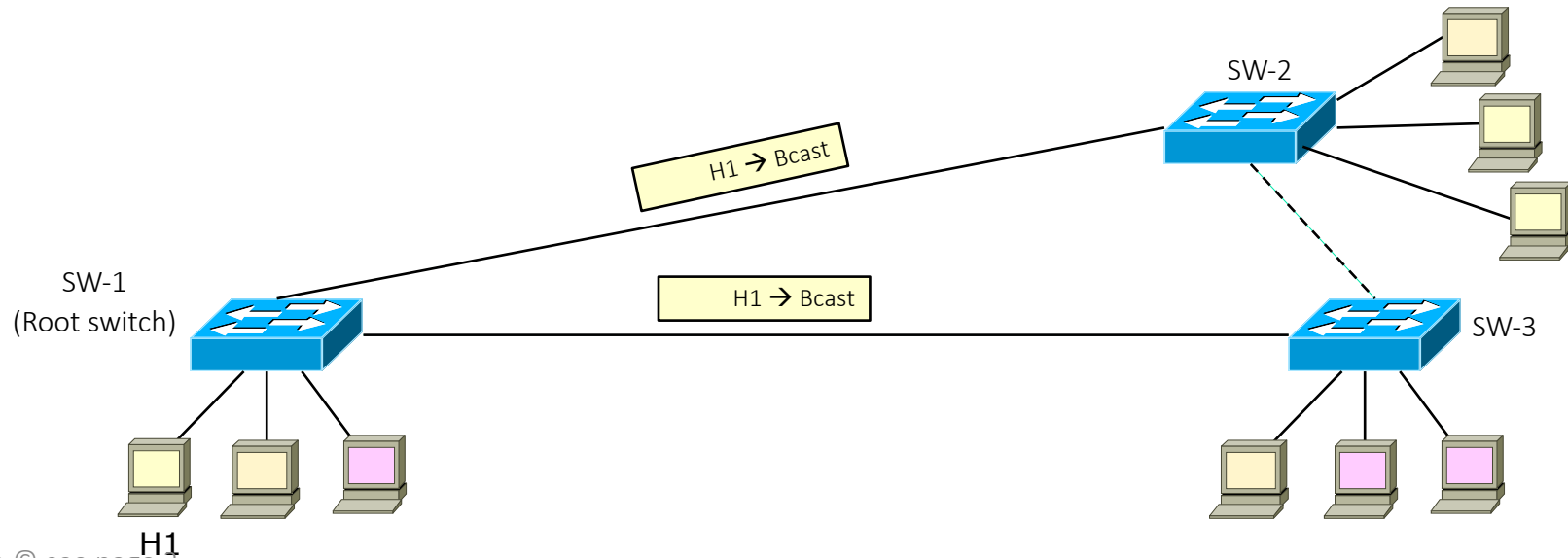
VLANs and Spanning Tree

- In theory, they are completely independent
 - First, Spanning Tree is computed in order to disable loops
 - Then, VLANs are used on the resulting topology
 - Unique forwarding tree for all the VLANs
- Almost all vendors offer Per-VLAN Spanning Tree
 - Most vendors can turn back to a unique STP via configuration
 - Cisco cannot (Per-VLAN STP is the only option)



VLANs and network isolation (1)

- Network isolation is not complete, even with VLANs
 - Although frames cannot cross the border of a VLAN, links are shared, hence a problem on a *link*, caused by the traffic of one VLAN, may affect other VLANs



VLANs and network isolation (2)

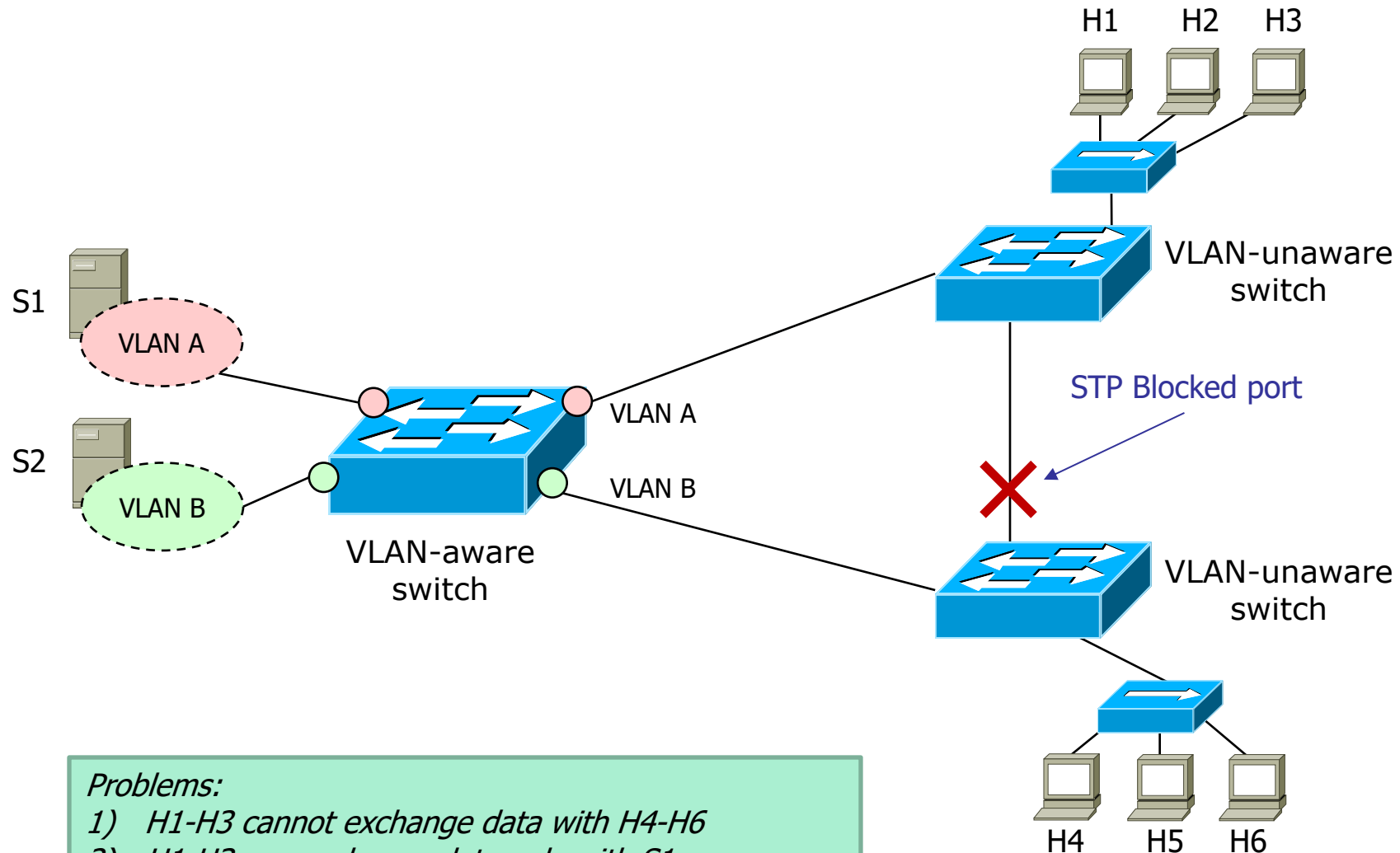
- For example, VLANs do not protect from broadcast storms
 - In fact, *broadcast traffic* is sent on the entire network
 - Except on the edge ports, since those are assigned to a specific VLAN
 - A trunk link may be saturated by a broadcast storm on a VLAN
 - Other VLANs do not receive that broadcast but...
 - ... the trunk link is congested and it may be unable to transport the traffic of other VLANs
- Per-VLAN QoS may be required
 - E.g., “Round-robin” service model based on VLAN ID, which guarantees a minimum amount of bandwidth to each VLAN



VLANs and network switches

- Two types of switches
 - VLAN-Aware: handle tagged and untagged frames
 - VLAN-Unaware: do not accept tagged frames
 - May discard frames (if too big)
 - Low-end devices
- Availability on the market
 - Almost all professional products can handle VLAN tagging
 - Almost all domestic products do not have VLAN support
- VLANs are no longer a “plug and play” technology
 - STP is (with some limitations)
 - This is one of the reasons VLANs are not supported on domestic switches
 - Typical users are not skilled enough to configure them

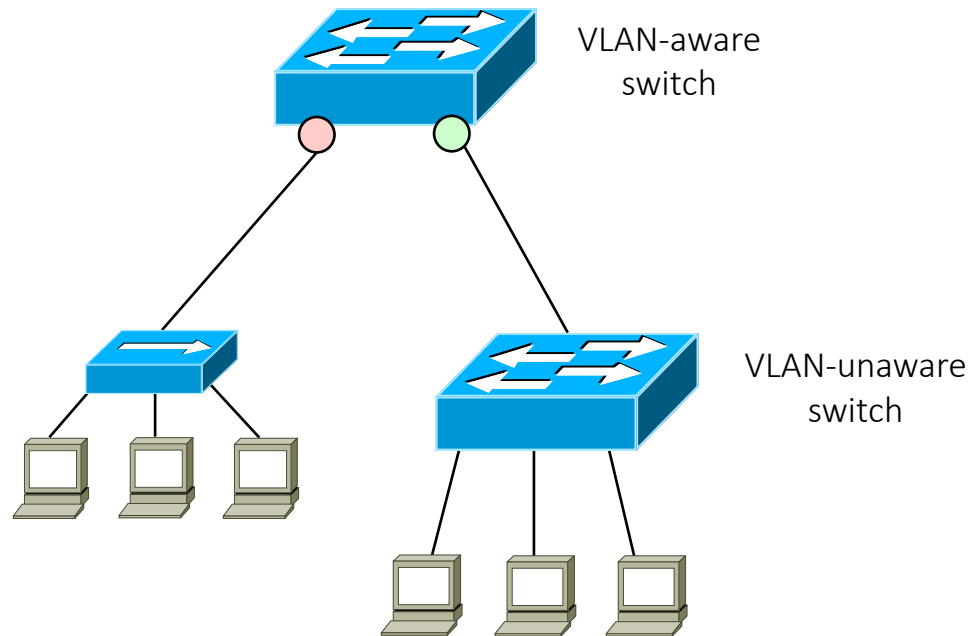
Mixing VLAN-aware/unaware switches (1)



Problems:

- 1) H1-H3 cannot exchange data with H4-H6
- 2) H1-H3 can exchange data only with S1
- 3) H4-H6 exchange data only with S2

Mixing VLAN-aware/unaware switches (2)



VLAN-unaware switches may be OK in the access side (e.g., in order to add new ports), provided that all clients belong to the same VLAN

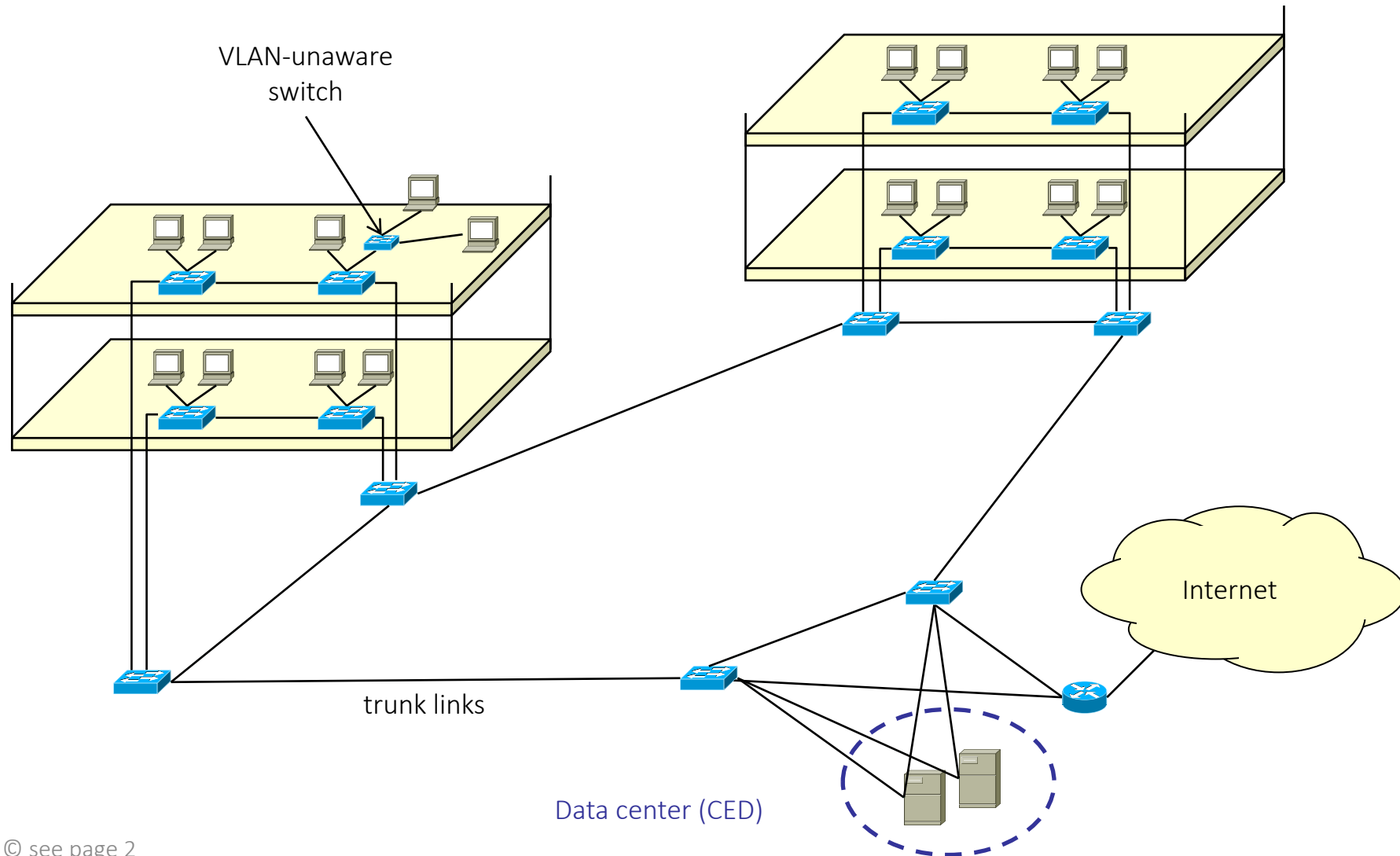
Corollary

It is pretty common to have VLAN-unaware switches in corporate networks.

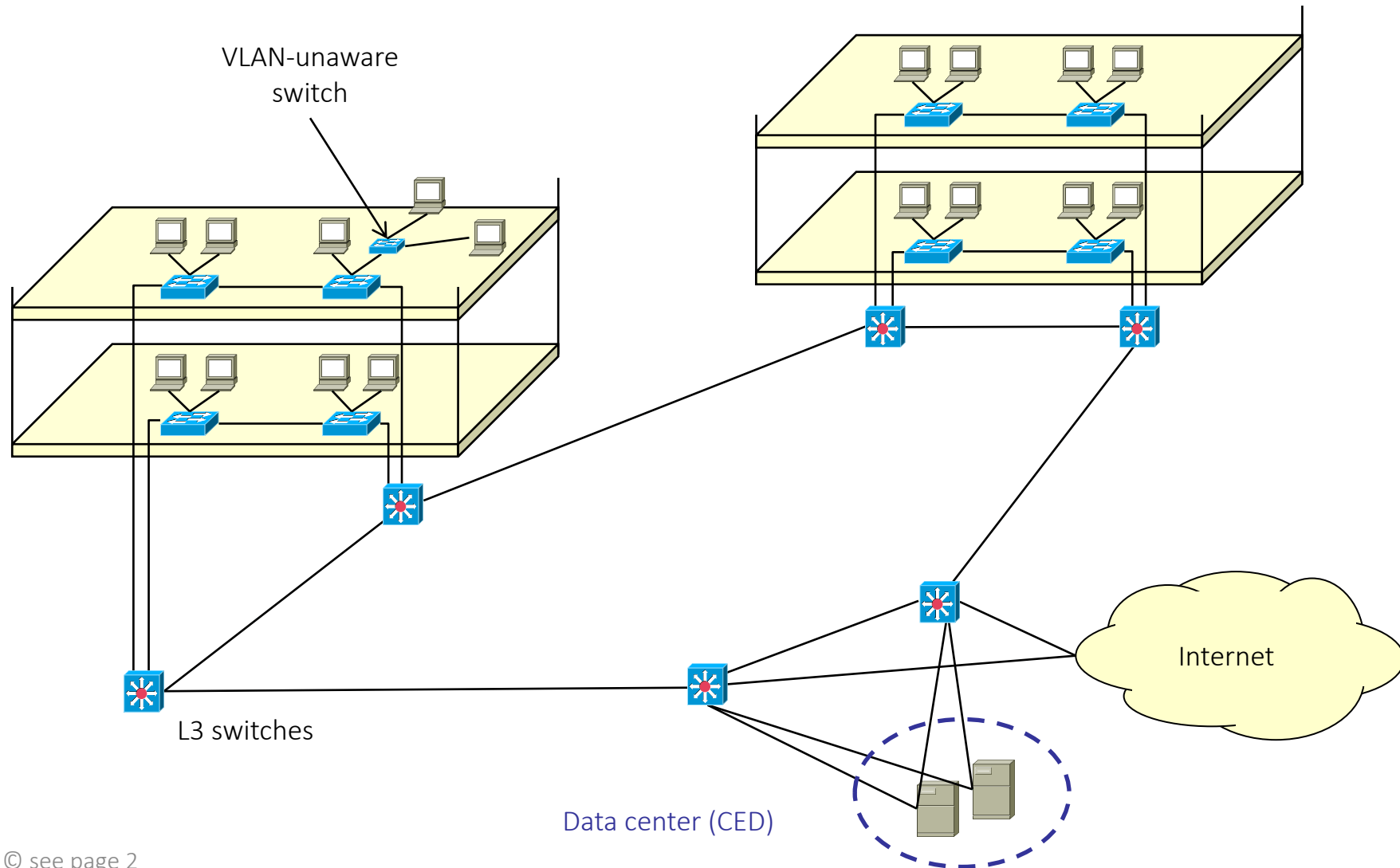
Network managers typically deploy only professional switches (with VLAN support) but often end users have some limitations (e.g., necessity to attach multiple hosts on a single network socket) and tend to sort those problems out by themselves, which usually means they buy the cheapest switch on the market, which does not have VLAN support.

Therefore, it is important that the network manager takes into account those situations (even if he does not know exactly where those switches may be installed) in order to prevent possible misbehaviour of the network.

Ethernet LAN design (1)



Ethernet LAN design (2)



Conclusions

- Modern wired LANs are based on Switched Ethernet
 - Star topology with full-duplex links
 - Collision domain no longer exists
 - No need for CSMA/CD
 - Fault tolerance given by redundancy + Spanning Tree Protocol
- Wide adoption of VLANs
 - Traffic isolation
 - Broadcast domain size reduction
 - Routers required for both internal and external communications
- L3 closer to the users
 - L3 switches

Annex 1: Configuring VLANs on Cisco switches

- VLAN creation

```
Switch# vlan database
Switch(vlan)#vlan 2 name Administration
VLAN 2 added:
  Name: Administration
```

```
Switch(vlan)#exit
APPLY completed.
Exiting....
switch#
```

Note: the command for adding an entry in the VLAN database changes according to the different version of the Cisco IOS and given device in use.

In more modern devices, the *vlan database* command can be issued also in standard configuration mode. In others (e.g. Cisco 6500) the command is even different.



Annex 1: Configuring VLANs on Cisco switches

- VLAN port association
 - Default behavior: a port is considered Access and associated to a default VLAN
 - The switch has a VLAN-unaware behavior

```
Switch# configure terminal
Switch(config)#interface FastEthernet 0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch# show vlan brief
```

	VLAN Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4
2	Administration	active	Fa0/1



Annex 1: Configuring VLANs on Cisco switches

- Configuration of the trunk port

```
Switch# configure terminal
Switch(config)# interface FastEthernet 0/2
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan
                        add 1,2 [or "all"]
Switch(config-if)# exit
Switch#
```

